2026 Level 2 - Portfolio Management

Economics and Investment Markets

a. explain the notion that to affect market values, economic factors must affect one or more of the following: 1) default-free interest rates across maturities, 2) the timing and/or magnitude of expected cash flows, and 3) risk premiums

b. explain the role of expectations and changes in expectations in market valuation

c. explain the relationship between the long-term growth rate of the economy, the volatility of the growth rate, and the average level of real short-term interest rates

d. explain how the phase of the business cycle affects policy and short-term interest rates, the slope of the term structure of interest rates, and the relative performance of bonds of differing maturities

e. describe the factors that affect yield spreads between non-inflation-adjusted and inflation-indexed bonds

f. explain how the phase of the business cycle affects credit spreads and the performance of credit-sensitive fixed-income instruments

g. explain how the characteristics of the markets for a company's products affect the company's credit quality

h. explain how the phase of the business cycle affects short-term and long-term earnings growth expectations

i. explain the relationship between the consumption hedging properties of equity and the equity risk premium

j. describe cyclical effects on valuation multiples

k. describe the economic factors affecting investment in commercial real estate

LOSs will match between the video and the PDFs, but may be in a different order than the CFAI readings

Economics & Investment Markets

Page 1
LOS a, b
– explain

investment markets

bonds
equities

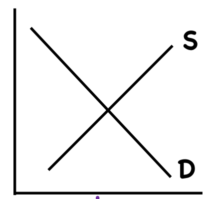
generates future growth and employment

$$\text{Investment} \xrightarrow{\text{invest}} \text{Production} \quad (S)$$

=

$$\text{Savings} \xleftarrow[\text{the future}]{\text{defer to}} \text{Consumption} \quad (D)$$

(present)



real economy

$$P_t^i = \sum_{s=1}^{N} \frac{E_t\left(\widetilde{CF}_{t+s}\right)}{\left(1 + I_{t,s} + \theta_{t,s} + \rho_{t,s}\right)^s}$$

compensation required → I real risk-free rate

→ θ expected inflation

→ ρ risk premium associated with $\widetilde{CF}$

$\widetilde{CF}_{t+s}$ – uncertain cash flows

$E_t[\widetilde{CF}_{t+s}]$ – expectation of CF conditional on information available today

Page 2
LOS a, b
– explain

- all financial instruments essentially represent claims on an underlying economy

$$P_t^i = \sum_{s=1}^{N} \frac{E_t\left(\widetilde{CF}_{t+s}\right)}{\underbrace{\left(1 + I_{t,s} + \theta_{t,s} + \rho_{t,s}\right)^s}}$$

num.
economy affects
den.

depends on future expected cash flows conditioned on current information

↓

reflects all expected/anticipated information

values vary over time as expectations regarding economic growth, inflation, and cash flow risk change

∴ unexpected information will (may) affect:

$E_t\left(\widetilde{CF}_{t+s}\right)$ $\quad I_{t,s}$ $\quad \theta_{t,s}$ $\quad \rho_{t,s}$

Page 3
LOS a, b
– explain

I/ real risk free rate (assume a 1 yr. zero, FV = $1)

current — future

t — $t + s$

C_t — has utility today

C_{t+s} — has future utility — consumption

$$\frac{U_{t+s}}{U_t} = \tilde{m}_{t,s}$$ – intertemporal rate of substitution

$(U_t > U_{t+s})$ – always

expected future conditions		$P_t^i = E(\tilde{m}_{t+s})$ e.g./	$I = \frac{i}{(\tilde{m}_{t,s})} - 1$
good	$U_{t+s} \downarrow$, $\tilde{m}_{t,s}$ lower	.92	$\left(1/.92 - 1\right) = 8.696\%$ (higher)
bad	$U_{t+s} \uparrow$, $\tilde{m}_{t,s}$ higher	.94	$\left(1/.94 - 1\right) = 6.38\%$ (lower)

Page 4
LOS a, b
– explain

I/ real risk free rate (assume a 1 yr. zero, FV = $1)

- in words
  - if future economic conditions are expected to be good, future consumption will be high, since future incomes will be high
    = less of a need to save today, so consuming today has much more value than consuming tomorrow (i.e. $\tilde{m}_{t,s} = {U_{t+s}}/{U_t}$ will be low)

Since $P_t^i = E_t(\tilde{m}_{t+s})$ = higher rates needed to induce savings

assume $\tilde{m}_{t,s}$ = .9400 — if $P_t^i < .9400$, rates are higher than required

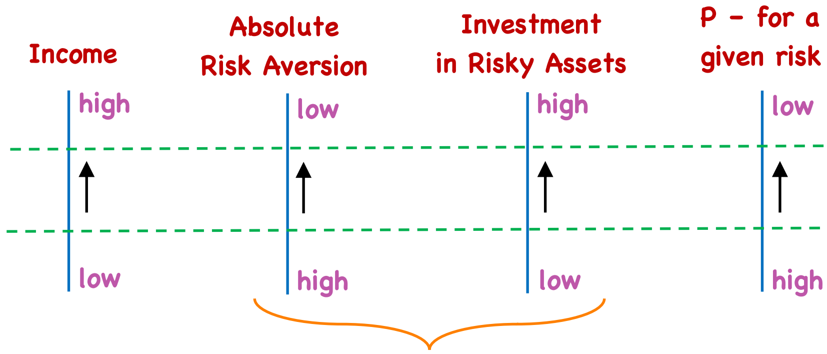
$C_t \downarrow$ $S \uparrow$ $\Rightarrow$ $P_t^i \uparrow$ & $I_{t,s} \downarrow$

$U_t \uparrow$, $\tilde{m}_{t,s} \downarrow$

move closer together

I/ real risk free rate

Page 5
LOS a, b
- explain



wealthier individuals are more willing to take on a given risk

- lowers the price of safe assets

∴ as incomes rise, equilibrium return on safe assets increase (i.e. price drops)

⇒ real risk-free rate increases

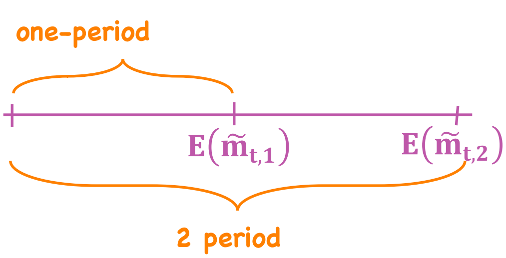
I/ real risk free rate

Page 6
LOS a, b
- explain

$$\Rightarrow P_t^i = \frac{1}{1 + I_{t,s}} = E(\tilde{m}_{t+s})$$ price of a one-period, \$1 payoff

$$\Rightarrow P_t^i = \frac{E[P_{t+1}^i]}{1 + I_{t,s}} - \underbrace{\text{risk premium}}_{\text{on a default-free bond?}}$$ two-period

one period
price of the price of a one-period, \$1 payoff



- more uncertainty in the expectation of the expected marginal utility two periods out

∴ real risk-free rate increases as time-to-maturity increases

Page 7
LOS a, b
- explain

- What have we said so far?
  1. Expectations about the future affect savings
  2. Savings affect prices of financial assets

good times ahead → higher current consumption = higher returns required to induce saving = lower prices for risk-free assets but/ higher prices for risky assets (less uncertainty about future cash flows)
(reverse for bad times)

Sell bonds/buy stocks

3. Longer time periods = more uncertainty about future conditions = higher returns required to induce saving
4. As incomes rise, returns on safe assets inadequate, ∴ more risky assets held

Economic Growth

Page 8
LOS c
- explain

⇒ Increase in real GDP growth ⇒ increase in real interest rate
(less saving, more borrowing)

(good times ahead, higher future incomes)

⇒ real interest rate (I) in economy with higher trend real GDP growth > I in economy with lower trend real GDP growth

= forecasts of GDP growth are $\underbrace{\text{more uncertain}}_{\text{risk premium required}}$ the more volatile GDP growth is

I higher in economy with more volatile GDP growth

Business Cycle

Page 9
LOS d, e
- explain

⇒ introduce inflation/

expected inflation = $\theta_{t,s}$

uncertainty in expected inflation = $\pi_{t,s}$ - over very short periods of time $\pi_{t,s} = 0$

$$\therefore \; P_t^i = \frac{CF_{t,s}^i}{(1 + \underbrace{I_{t,s} + \theta_{t,s} + \pi_{t,s}}_{\text{nominal interest rate}})^s} \quad \text{(T-Bill)}$$

⇒ $\underbrace{\text{T-Bill}}_{\text{often used to implement monetary policy}}$ rates & the Business Cycle/

often used to implement $\underbrace{\text{monetary policy}}$ (open-market operations)

r = I + θ

- cut policy rates when business activity and/or inflation judged to be too low
- raising rates when the opposite is the case

Page 10
LOS d, e
- explain

⇒ T-Bill rate & the Business Cycle/

policy rate = I + θ + .5(θ − θ*) + .5(y − y*)

real | inflation | inflation gap | output gap

$\underbrace{\text{inflation} + \text{inflation gap}}_{1.5\theta \,-\, .5\theta^*}$

$\underbrace{1.5\theta - .5\theta^*}_{\text{larger weight on inflation}}$

⇒ Longer-term bonds & the Business Cycle/

- the greater the uncertainty about the real value or the payoff (i.e. lower confidence in the ability to form views about future inflation) will cause investors to demand a premium ($\pi_{t,s}$)

BEI = r − I
break-even inflation rate

(yield on a zero nominal bond − yield on a zero real)
default free

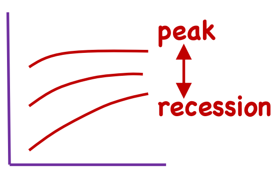
Page 11
LOS d, e
– explain

⇒ Longer-term bonds & the Business Cycle/

- evolution of I over time driven mainly by $\tilde{m}_{t,s}$
- evolution of r over time will also be driven by
  ① changing expectations about inflation, and
  ② changing perceptions about the uncertainty of the future inflation environment

∴ BEI reflect both ① & ②

⇒ Default-Free Yield Curve & the Business Cycle



- short-end ⇒ policy impact
- full curve ⇒ expected impact of growth
  ⇒ levels of expected future inflation

Credit Risk

Page 12
LOS f, g
– explain

$$P_t^i = \sum_{s=1}^{N} \frac{E_t[\widetilde{CF}_{t+s}^i]}{(1 + I_{t,s} + \theta_{t,s} + \pi_{t,s} + \gamma_{t,s})^s}$$

credit premium

- credit spread = corporate 10 yr. – government 10 yr. (e.g.)
- tends to widen (i.e. the credit premium increases) in times of economic weakness
  - spreads of lower-rated bonds more volatile
    i.e. contract more in good times
    widen more in bad times
    } versus highly-rated bonds
- some sectors are more sensitive to the business cycle than others (i.e. will have more volatile spreads)

cyclical (discretionary) vs. non-cyclical (defensive)

⇒ Company specific factors/

Page 13
LOS f, g
- explain

$\underbrace{\text{• profitable, low debt interest payments, low debt levels}}_{\text{higher credit ratings}}$

Equities

LOS h, i
- explain

$$P_t^i = \sum_{s=1}^{\infty} \frac{E_t[\widetilde{CF}_{t+s}^i]}{(1 + l_{t,s} + \theta_{t,s} + \pi_{t,s} + \gamma_{t,s}^i + \kappa_{t,s}^i)^s}$$

additional return required to invest in equities over and above credit risky bonds
(equity premium relative to credit risky bonds)

$$(\gamma_{t,s}^i + \kappa_{t,s}^i) = \lambda_{t,s}^i$$

↓

ERP

Page 14
LOS h, i
- explain

• in terms of deferred consumption, the ERP compensates for the poor consumption hedging properties of equities (low/no payoff in bad times)
- sharp falls in equities are associated with recessions

⇒ Earnings Growth & The Business Cycle/
- declines associated with sharp declines in real earnings
- sharp increase in profit growth occur at the end of a period of recession (leading indicator)

- relationship of earnings with the business cycle also dependent on the type of product/service sold

non-cyclical/defensive	cyclical/discretionary
less sensitive	more sensitive

⇒ ERP/ – since equities are a poor hedge against bad consumption outcomes

⇒ ERP is positive

Page 15
LOS h, i
– explain

⇒ Valuation Multiples/ P/E or P/B (e.g.)

higher forward P/E's → an increase in $E_t[\widetilde{CF}^i_{t+s}]$
→ a fall in $I_{t,s}$
→ a fall in $\theta_{t,s}$
→ a fall in $\pi_{t,s}$
→ a fall in $\lambda^i_{t,s}$

LOS j
– describe

⇒ Investment Strategy/ growth vs. value

– value tends to outperform growth in the aftermath of a recession
– growth tends to outperform when the economy is expanding

LOS k
– describe

– small cap, mid-cap, large-cap

small cap – larger ERP
large-cap – smaller ERP

Page 16
LOS k, l
– describe

Commercial Real Estate

$$P^i_t = \sum_{s=1}^{\infty} \frac{E_t[\widetilde{CF}^i_{t+s}]}{(1 + I_{t,s} + \theta_{t,s} + \pi_{t,s} + \gamma^i_{t,s} + \kappa^i_{t,s} + \phi^i_{t,s})^s}$$

illiquidity

– cash flows from rents – low sensitivity to business cycle

– property values – high sensitivity

Analysis of Active Portfolio Management

a. describe how value added by active management is measured

b. calculate and interpret the information ratio (ex post and ex ante) and contrast it to the Sharpe ratio

c. describe and interpret the fundamental law of active portfolio management, including its component terms–transfer coefficient, information coefficient, breadth, and active risk (aggressiveness)

d. explain how the information ratio may be useful in investment manager selection and choosing the level of active portfolio risk

e. compare active management strategies, including market timing and security selection, and evaluate strategy changes in terms of the fundamental law of active management

f. describe the practical strengths and limitations of the fundamental law of active management

Active Management

Page 1
LOS a
– describe

- objective of active management – add value by doing better than a benchmark portfolio (relative performance comparison)
- Choice of Benchmark/
  1) representative of the assets from which the investor will select
  2) positions in the benchmark can actually be replicated at low cost
  3) benchmark weights are verifiable ex ante (before the fact), and return data are timely ex post (after the fact)

$$R_B = \sum_{i=1}^{N} w_{B_i} R_i \qquad R_P = \sum_{i=1}^{N} w_{P_i} R_i$$

N = # of securities

Page 2
LOS a
– describe

active return → $R_A = R_P - R_B$

$$\Rightarrow R_A = \sum_{i=1}^{N} w_{P_i} R_i - \sum_{i=1}^{N} w_{B_i} R_i$$

$$= \sum_{i=1}^{N} (w_{P_i} - w_{B_i}) R_i$$

$$= \sum_{i=1}^{N} \Delta w_i R_i$$

(ΔW_i = active weights)

alpha → $\alpha_P = R_P - \beta R_B$
(risk adjusted active return)

– extra for understanding/

since $\Sigma w_P = 1$ & $\Sigma w_{B_i} = 1$ then $\underbrace{\Sigma(w_{P_i} - w_{B_i}) = 0}_{\Sigma \Delta w_i = 0}$

so...

$$R_A = \Sigma \Delta w_i R_i - \underbrace{\Sigma \Delta w_i R_B}_{= 0 \text{ since } \Sigma \Delta w_i = 0 \text{ and } R_B = \text{constant}}$$

$$\therefore R_A = \Sigma \Delta w_i (R_i - R_B)$$

$$= \Sigma \Delta w_i R_{A_i}$$

value added is the sum of the product of the active weights and active security returns

$$R_A = \sum_{i=1}^{N} \Delta w_i R_{A_i} = \sum_{i=1}^{N} \Delta w_i \underbrace{(R_i - R_B)}$$

Page 3
LOS a
– describe

positive value is added under 2 conditions

① securities in which $R_i > R_B$ are overweighted
② securities in which $R_i < R_B$ are underweighted

e.g./	B	P
stocks	60%	70%
bonds	40%	30%

w_B & w_P

R_B = .6(.14) + .4(.02) = 9.2%
R_P = .7(.14) + .3(.02) = 10.4%

$R_A = R_P - R_B = 1.2\%$

$R_{stocks} = 14\%$

$R_{bonds} = 2\%$

wrong analysis: $R_{P_{stocks}} - R_{B_{stocks}} = +1.4\%$

$R_{P_{bonds}} - R_{B_{stocks}} = -\ .2\%$

1.2%

correct analysis:

$(R_{stocks} - R_B)\ \Delta w_{stocks} = (.14 - .092)(.10) = .0048$

$(R_{bonds} - R_B)\ \Delta w_{bonds} = (.02 - .092)(-.10) = .0072$

$.012 \text{ or } 1.2\%$

Country	Benchmark Weight	Portfolio Weight	2013 Return
United Kingdom	22%	16%	20.7%
Japan	21%	14%	27.3%
France	10%	8%	27.7%
Germany	9%	24%	32.4%
Other Countries	38%	38%	18.8%

Page 4
LOS a
– describe

$$R_A = \sum_{i=1}^{N} \Delta w_i R_i$$

$= (.16 - .22).207$
$+ (.14 - .21).273$
$+ (.08 - .10).277$
$+ (.24 - .09).324$
$= -.01242 - .01911$
$- .00554 + .0486$
$= .01153$
or
1.153%

$R_B = .22(.207) + .21(.273) + .10(.277) + .09(.324)$
$+ .38(.188) = .23117 \text{ or } \underline{23.117}\%$

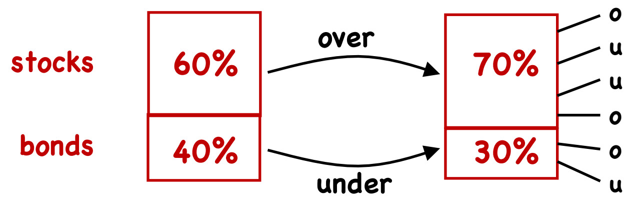
$R_A = \Sigma \Delta w_i (R_i - R_B)$
= (-.06)(.207 - .23117)
+ (-.07)(.273 - .23117)
+ (-.02)(.277 - .23117)
+ (.15)(.324 - .23117)

= .0014502 (+)
- .0029281 (-)
- .0009166 (-)
+ .0139245 (+)
= .01153 or 1.153%

Page 5
LOS a
- describe

⇒ Decomposition of Value Added

due to asset allocation — due to security selection



M asset classes
N securities

$$R_A = \sum_{j=1}^{M} w_{P_j} R_{P_j} - \sum_{j=1}^{M} w_{B_j} R_{B_j}$$ total value added

$$= \sum_{j=1}^{M} \Delta w_j R_{B_j} + \sum_{j=1}^{M} w_{P_j} R_{A_j}$$

$$= \left(\Delta w_{stocks} R_{B_{stocks}} + \Delta w_{bonds} R_{B_{bonds}}\right)$$ asset allocation

$$+ \left(w_{P_{stocks}} R_{A_{stocks}} + w_{P_{bonds}} R_{A_{bonds}}\right)$$ security selection

Page 6
LOS a
- describe

$$= \left(\Delta w_{stocks} R_{B_{stocks}} + \Delta w_{bonds} R_{B_{bonds}}\right)$$ asset allocation

$$+ \left(w_{P_{stocks}} R_{A_{stocks}} + w_{P_{bonds}} R_{A_{bonds}}\right)$$ security selection

e.g./

	B	P
stocks	15%	17%
bonds	4%	4.2%
	(60/40)	(70/30)

R_B = .6(.15) + .4(.04) = 10.6%

R_P = .7(.17) + .3(.042) = 13.16%

R_A = R_P − R_B = 2.56%

① What if the benchmark were 70/30?

.7(.15) + .3(.04) = 11.7%

versus .6(.15) + .4(.04) = 10.6%

.1(.15) + (−.1)(.04) = 1.1%

$$R_A = \left[\sum_{j=1}^{M} \Delta w_j R_{B_j}\right]$$

② Now account for better performance

.7(.02) + .3(.002) = .0146

(1.1% + 1.46% = 2.56%)

$$+ \left[\sum_{j=1}^{M} w_{P_j} R_{A_j}\right]$$

Fund	Fund Return (%)	Benchmark Return (%)	Value Added (%)
Fidelity Magellan (.68)	35.3	(.60) 32.3	3.0
PIMCO Total Return (.32)	−1.9	(.40) −2.0	0.1
Portfolio Return	23.4	18.6	4.8
	R_P	R_B	R_A

Page 7
LOS a
- describe

$R_P = .68(.353) + .32(-.019)$
$= .23396$

asset allocation/
Σ(active weights × R_B)
$= .08(.323) + (-.08)(-.02)$
$= .02584 + .0016$
$= .02744$

$R_B = .6(.323) + .4(-.02)$
$= .1858$

security selection/
Σ(portfolio weights × $R_{A,j}$)
$= .68(.03) + .32(.001)$
$= .0204 + .00032$
$= .02072$

$.02744 + .02072 = .04816$ or 4.816%

Information Ratio

Sharpe Ratio

$$SR_P = \frac{R_P - R_F}{\sigma_{R_P}}$$

- absolute expected or realized reward/risk measure

Page 8
LOS b
- calculate
- interpret
- contrast

- unaffected by the additions of cash or leverage/

w_P - weight on actively managed portfolio
$(1 - w_P)$ - weight on cash

$$R_C = w_P R_P + (1 - w_P) R_F$$

$$SR_C = \frac{R_C - R_F}{\sigma_{R_C}} = \frac{w_P R_P + R_F - w_P R_F - R_F}{w_P \sigma_{R_P}} = \frac{w_P R_P - w_P R_F}{w_P \sigma_{R_P}}$$

$$= \frac{w_P (R_P - R_F)}{w_P \sigma_{R_P}} = \frac{R_P - R_F}{\sigma_{R_P}}$$

e.g./ Fund $R_P = 10.4\%$

$\sigma_{R,p} = 17.3\%$

$R_F = 2.8\%$

$$SR_p = \frac{.104 - .028}{.173} = .4393$$

Page 9
LOS b
- calculate
- interpret
- contrast

- reduce $\sigma_{R,p}$ to 10% $\quad w_P = .10/.173 = .578$

$(1 - w_P) = .422$ held in cash

$$R_C = .578(.104) + .422(.028) = .060112 + 0.011816 = .071928$$

$$SR_C = \frac{.071928 - .028}{.578(.173)} = \frac{.043928}{.10} = .43928$$

e.g. 2/

	Large Cap.	Small Cap.
E(R)	10%	13.4%
E(σ)	15.2%	21.1% → reduce to 15.2%
SR	.47	.50

$$.152/.211 = .7204 \text{ (Sm. Cap.)}$$

$$.2796 \text{ (Cash)}$$

$$R_C = .7204(.134) + .2796(.028)$$

$$= .1044$$

$$SR_C = \frac{.1044 - .028}{.7204(.211)} = \underline{.5026}$$

⇒ Information Ratio/

$$IR = \frac{R_P - R_B}{\sigma_{(R_P - R_B)}} = \frac{R_A}{\sigma_{R_A}} \quad \left(\frac{\text{active return}}{\text{active risk}}\right)$$

Page 10
LOS b
- calculate
- interpret
- contrast

Passive Fund $\quad$ Benchmark Fund

$$\underbrace{SR_F \quad = \quad SR_B}_{\text{no active positions}}$$

IR = 0

IR unaffected by the aggressiveness of active weights

Market Neutral Fund (β = 0)

$$\underbrace{SR_P = IR_P}_{\text{all active}}$$

Risk-Free Asset (β = 0)

$$R_C = \sum_{i=1}^{N} c\Delta w_i R_{A_i}$$

$$= c \sum_{i=1}^{N} \Delta w_i R_{A_i}$$

$$= cR_A$$

$$IR = \frac{cR_A}{c\sigma_{R_A}} = \frac{R_A}{\sigma_{R_A}}$$

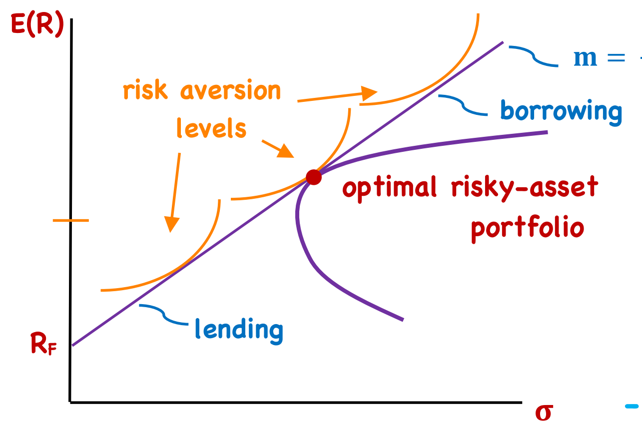
⇒ IR affected by cash/leverage

P + cash

IR ↓

Page 11
LOS b
- calculate
- interpret
- contrast

⇒ Recall from Level 1/



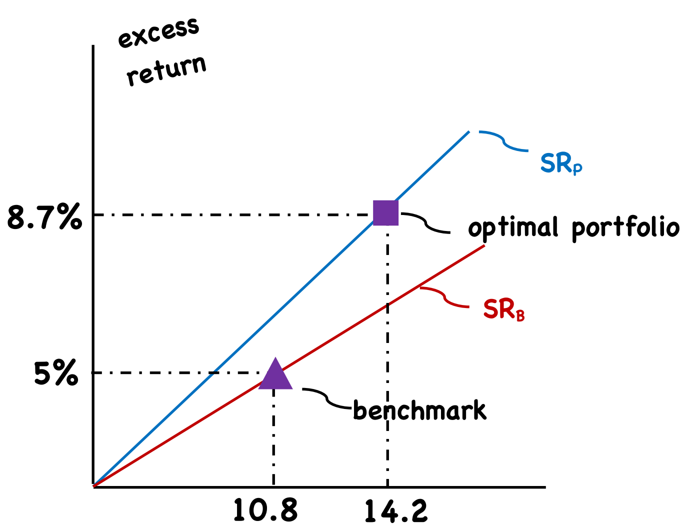
$$m = \frac{R_P - R_F}{\sigma_p} = SR_P$$

Passive/ risk/return can be adjusted with either cash (lending) or leverage (borrowing)

What about active investing?
- is there an optimal amount of active risk?

- adjusting active risk may involve investing in both the actively managed portfolio & the benchmark portfolio, but more likely modifying active weights

Page 12
LOS b
- calculate
- interpret
- contrast



- property of active management theory

$$SR_P^2 = SR_B^2 + IR^2$$

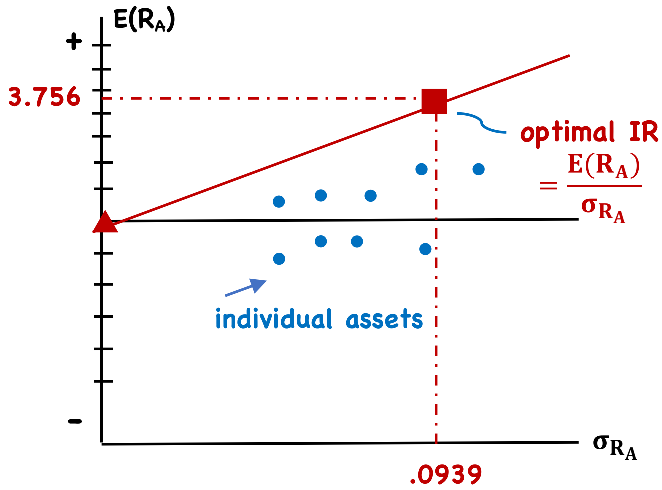
implies that the active portfolio with the highest IR will also have the highest SR

$SR_B = .05/.108 = .46 \quad SR_P = .087/.142 = .61$

- an optimal amount of active risk

$$\underbrace{\sigma_{R_A}}_{\text{aggressiveness (on the weightings)}} = IR/SR_B \cdot \sigma_{R_B}$$

Page 13
LOS b
- calculate
- interpret
- contrast



$$SR_P^2 = SR_B^2 + IR^2$$

$$.61^2 = .46^2 + IR^2$$

$$IR^2 = .61^2 - .46^2$$

$$\text{optimal IR} = \sqrt{.1605} = 0.400$$

$$\underbrace{\sigma_{R_A} = \frac{IR}{SR_B} \cdot \sigma_{R_B}}$$

$$\text{optimal } \sigma_{R_A} = \frac{.40}{.46}(.108) = .0939$$

and since $IR = \frac{E(R_A)}{\sigma_{R_A}}$

$$.40 = \frac{E(R_A)}{.0939}$$

$$E(R_A) = 3.756$$

- weights on individual securities/asset classes can be adjusted to get $\sigma_{R,A}$ to 9.39%

or/

- combine the optimal portfolio and the benchmark portfolio

The Fundamental Law

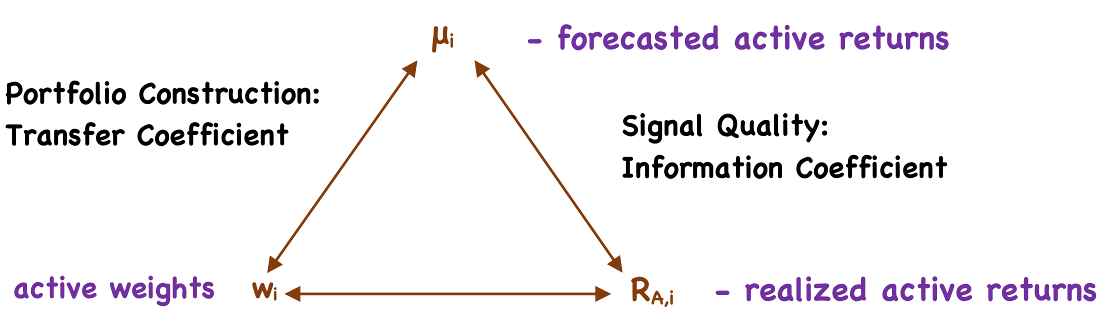
Page 14
LOS c
- state
- interpret

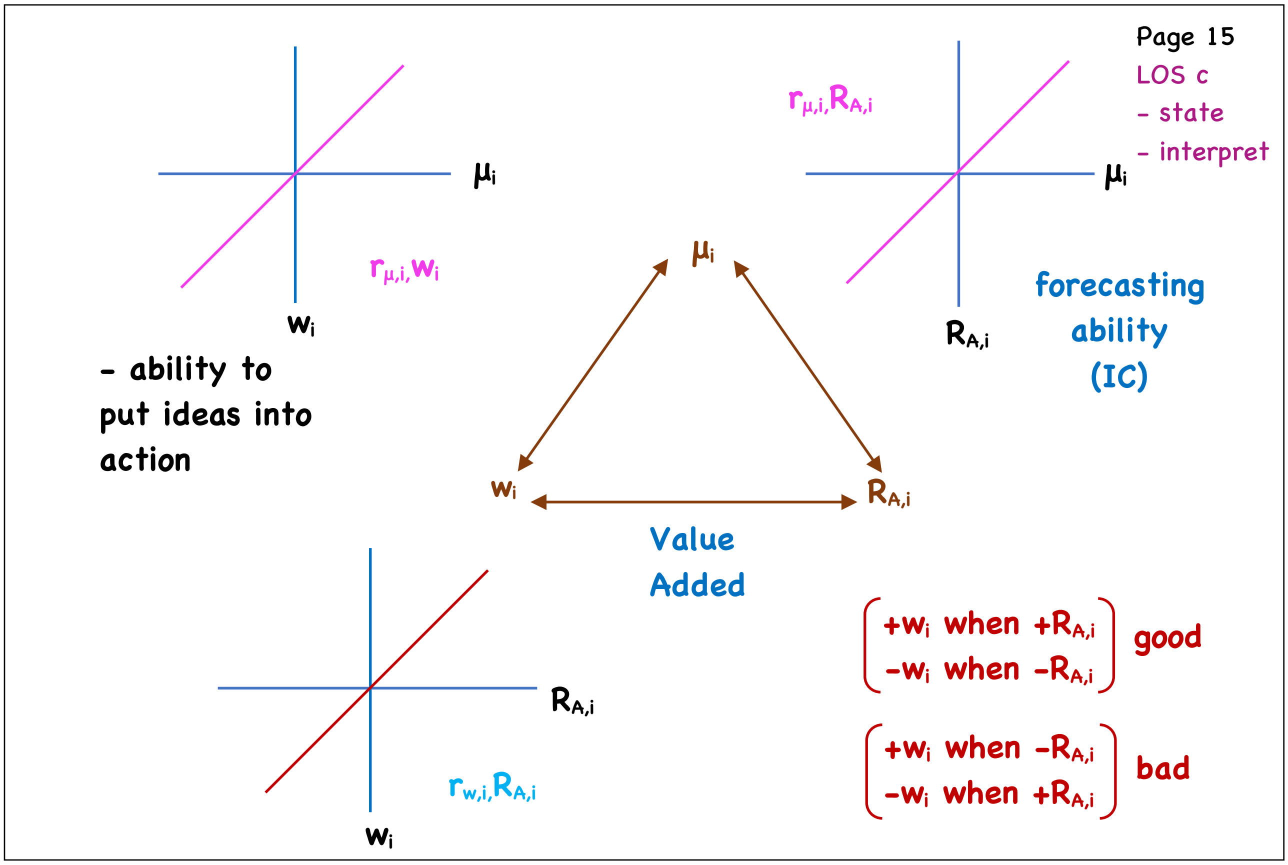
- investor is concerned about maximizing the managed portfolio's active return, subject to a limit on $\underbrace{\text{active risk}}_{\text{benchmark tracking risk}}$

- Investor Forecasts

$R_{A,i} = R_i - R_B$ for N individual assets

let $\mu_i = E(R_{A,i})$ – the investor's subjective expectation of the active return on asset i





Page 16
LOS c
- state
- interpret

$$\Delta w_i^* = \frac{\mu_i}{\sigma_i^2} \frac{\sigma_A}{\sqrt{\sum_{i=1}^{N} \frac{\mu_i^2}{\sigma_i^2}}}$$

- the mean-variance optimal active security weights for uncorrelated active returns, subject to a limit on active risk

where $\mu_i = IC\sigma_i S_i$

IC: E(IC) (positive)

σ_i: (positive)

S_i: standardized forecasts of E(R_i) (scores) (based on overperformance (+), neutral (o), or underperformance (-))

$\Sigma S_i = 0$

$\sigma_S = 1$

$$\therefore \Delta w_i^* = \frac{\mu_i}{\sigma_i^2} \cdot \frac{\sigma_A}{IC\sqrt{BR}}$$

breadth $\sim \sqrt{N}$

- more accurately, the # of independent decisions made/yr. in constructing the portfolio

IC: $\text{corr.}\left(\frac{R_{A_i}}{\sigma_i}, \frac{\mu_i}{\sigma_i}\right)$

risk adjusted

Page 17
LOS c
- state
- interpret

e.g./

Security	Score	Volatility
1	1.0	25%
2	1.0	50%
3	-1.0	25%
4	-1.0	50%

Securities 1–2: outperform
Securities 3–4: underperform

assume IC = .20

① Forecasted Active Returns/ $E(R_{A,i}) = \mu_i$

$\mu_i = IC\sigma_i S_i$

1 .20 (.25) (1) = 5%
2 .20 (.50) (1) = 10%
3 .20 (.25) (-1) = -5%
4 .20 (.50) (-1) = -10%

② Active returns are uncorrelated
- independent forecasts are made each yr.

∴ BR = 4

③ Max. $E(R_A)$ s.t. $\sigma_A = 9.0\%$
- calculate Δw_i^*

$$\therefore \Delta w_i^* = \frac{\mu_i}{\sigma_i^2} \cdot \frac{\sigma_A}{IC\sqrt{BR}}$$

$$\Delta w_1 = \left({.05}/{.25^2}\right)\left({.09}/{.20\sqrt{4}}\right) = 0.18$$

$$\Delta w_2 = \left({.10}/{.5^2}\right).225 = .09$$

$$\Delta w_3 = \left({-.05}/{.25^2}\right).225 = -.18$$

$$\Delta w_4 = \left({-.10}/{.5^2}\right).225 = -.09$$

$$\sum = 0$$

Page 18
LOS c
- state
- interpret

⇒ The Basic Fundamental Law/

$$E(R_A)^* = IC\sqrt{BR}\,\sigma_A$$

* indicates the use of optimal active security weights

- optimal expected active return is a positive function of
  - forecasting ability (IC)
  - breadth
  - active risk

Recall: $IR = \frac{E(R_A)}{\sigma_A}$

So… $IR^* = \frac{IC\sqrt{BR}\,\sigma_A}{\sigma_A} = IC\sqrt{BR}$ thus $E(R_A)^* = IR^*\,\sigma_A$

e.g. revisited

Security	Δw_i	μ_i	w_{B_i}	w_{P_i}	R_{P_i}
1	.18	.05	.25	.43	.15
2	.09	.10	.25	.34	.20
3	-.18	-.05	.25	.07	.05
4	-.09	-.10	.25	.16	.0

$$E(R_P) = .43(.15) + .34(.20) + .07(.05) + .16(0) = 13.6\%$$

$$E(R_A) = 13.6\% - 10\% = 3.6\%$$

$$E(R_A) = .18(.05) + .09(.10) - .18(-.05) - .09(-.10) = 3.6\%$$

Page 19
LOS c
- state
- interpret

⇒ The Basic Fundamental Law/

e.g. revisited

Security	Δw_i	μ_i	w_{B_i}	w_{P_i}	R_{P_i}
1	.18	.05	.25	.43	.15
2	.09	.10	.25	.34	.20
3	-.18	-.05	.25	.07	.05
4	-.09	-.10	.25	.16	.0

$E(R_P) = 13.6\%$
$E(R_A) = 3.6\%$

$$\sigma^2_{R_A} = (.18)^2(.25)^2 + (.09)^2(.50)^2 + (-.18)^2(.25)^2 + (-.09)^2(.5)^2$$

$$\sigma_{R_A} = \sqrt{.002025 + .002025 + .002025 + .002025}$$

$$= .09$$

$\sigma_A = 9\%$

- using Basic Fundamental Law

$$E(R_A)^* = IC\sqrt{BR}\sigma_A = IR^*\ \sigma_A$$

$$= .20\sqrt{4}(.09) = .40(.09) = 3.6\%$$

Page 20
LOS c
- state
- interpret

⇒ The Full Fundamental Law/

$$TC = CORR\left(\frac{\mu_i}{\sigma_i}, \Delta w_i \sigma_i\right)$$

actual weights (for a constrained portfolio)

or

$$TC = CORR(\Delta w_i^* \sigma_i, \Delta w_i \sigma_i)$$

optimal

(if we assume all individual securities have the same σ, the correlations do not have to be risk-adjusted/weighted)

$\therefore\ E(R_A) = (TC)(IC)\sqrt{BR}\ \sigma_A$ ⇐ the Fundamental Law

$IR = (TC)(IC)\sqrt{BR}$ ⟵ $\left[IR = \frac{E(R_A)}{\sigma_A} = \frac{(TC)(IC)\sqrt{BR}\ \sigma_A}{\sigma_A}\right]$

Page 21
LOS c
- state
- interpret

e.g./

Security	$E(R_{A_i})$	σ_{A_i}	Δw_i^*	Δw_i
1	5%	25%	18%	6%
2	10%	50%	9%	4%
3	-5%	25%	-18%	7%
4	-10%	50%	-9%	17%

$\underbrace{\sigma_A = .09}_{\text{active risk constraint}}$

① $TC = CORR\left(\frac{\mu_i}{\sigma_i}, \Delta w_i \sigma_i\right) \longrightarrow$

Pairs

$.05/.25 = .20$	&	$.06(.25) = 0.015$
$.10/.50 = .20$		$.04(.50) = 0.02$
$-.05/.25 = -.20$		$.07(.25) = 0.0175$
$-.10/.50 = -.20$		$-.17(.50) = -0.085$

② $E(R_A)^* = 3.6\%$ $\quad \sigma_A^* = 9.0\%$

$E(R_A) = .06(.05) + .04(.10)$

$+ .07(-.05) - .17(-.10) = 2.1\%$

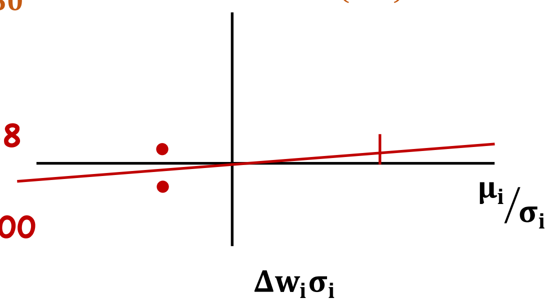
$\sigma_A = 9.0\%$

③ $E(R_A) = TC \cdot IC \cdot \sqrt{BR} \cdot \sigma_A$

$= (.58)(.20)(\sqrt{4})(.09)$

$= 2.088\%$

TC = .58
vs.
TC* = 1.00



Page 22
LOS c, d
- state
- interpret
- explain

- we can adjust active risk and return by investing in both the actively managed and benchmark portfolios

$\sigma_{R_A} = TC \frac{IR^*}{SR_B} \cdot \sigma_{R_B}$ - optimal amount for active risk

$\left(\sigma_{R,A} = \frac{IR}{SR_B} \cdot \sigma_{R_B} \text{ for unconstrained portfolios}\right)$

$SR_P^2 = SR_B^2 + (TC)^2 \underbrace{(IR^*)}^2$ $\quad$ $(\text{vs. } SR_P^2 = SR_B^2 + IR^2)$

the active portfolio with the highest IR will also have the highest SP_P

- E(IR) is the single best criterion for assessing active performance among actively managed funds with the same benchmark
- for any given asset class, an investor should choose the manager with the highest IR

Page 23
LOS c, d
- state
- interpret
- explain

$$\sigma_{R_A} = TC\,\frac{IR^*}{SR_B}\cdot\sigma_{R_B} \qquad SR_P^2 = SR_B^2 + (TC)^2(IR^*)^2$$

e.g./ $IR^* = .30$ $SR_B = 0.40$ $\sigma_{R,B} = .16$ $TC = .50$

optimal amount of aggressiveness/

$$\sigma_{R_A} = (.50)\frac{.30}{.40}(.16) = 6\%$$

then $SR_P = \sqrt{(64)^2 + (.5)^2(.3)^2} = 0.43$

- if the constrained portfolio had $\sigma_{R,A} = .08$

$$w_B = 1 - \frac{\text{optimal } \sigma_A}{\text{actual } \sigma_A} = 1 - {.06}/{.08} = .25$$

$$w_P = .75$$

Ex- Post Performance

Page 24
LOS c, d
- state
- interpret
- explain

$$E(R_A |\, IC_R) = (TC)(IC_R)\sqrt{BR}\,\sigma_A$$

↓

constraints

IC – expected forecasting ability
IC_R – realized (ex-post)

realized R_A = $E(R_A \mid IC_R)$ + Noise

skill + constraints – may be pos. or neg.

→ may be pos. or neg.

realized $\sigma_A^2 = (TC^2)\%$ $[(1-TC)^2]\%$

e.g./ BR = 100
IC = .05
TC = .80
σ_A = 4.0%

$$E(R_A) = TC\,(IC)\sqrt{BR}\,\sigma_A$$
$$= .80(.05)\sqrt{100}(.04) = 1.6\%$$

if IC_R = -.10

$$E(R_A |\, IC_R) = TC(IC)\sqrt{BR}\,\sigma_A$$
$$= .8(-.10)\sqrt{100}(.04) = -3.2\%$$

if R_A = -2.6%,
noise = +.6%

σ_A attributed to σ_{IC} = TC^2 = $.8^2$ = 64%

36% attributed to noise

Active Management Strategy

Page 25
LOS e
- compare
- evaluate

cross
sectional
N Securities ⟶ BR = N

Equities

t = 0 — — — — t = T

E —— B —— B —— E ⇒ marking timing (BR = T)

Fixed Income

t = 0, 1, 2, 3, t = T = 4

N Securities ↑ ↑ ↑ ↑ time series

BR = (N × T)

$$E(R_A) = TC(IC)\sqrt{BR}\,\sigma_A$$

$$IR = TC(IC)\sqrt{BR}$$

for an unconstrained portfolio (i.e. TC = 1), IR is invariant to changes in σ_A

- for a constrained portfolio, as σ_A increases, IR ↓ (constraints become more binding)

as σ_A ↑, aggressiveness ↑, $|\Delta w_i^*|$ ↑ & $TC = corr(\Delta w_i^*\,\sigma_i,\ \Delta w_i\,\sigma_i)$ ↓

Limitations

Page 26
LOS f
- describe

- ex-ante measurement of skill/
  - ⇒ IC
    - investor's tend to overestimate their skill
    - forecasting ability differs among different asset segments, and over time
- independence of Investment Decisions/ (lack of)
  - non-independence can result in very large BR, and large R_A vs σ_A (i.e. arbitrage, using derivatives)
  - fixed-income portfolios – bond returns much more highly correlated
  - time series – increasing the rebalancing frequency increases BR & IR, but decisions are less likely to be independent

Exchange-Traded Funds: Mechanics and Applications

a. explain the creation/redemption process of ETFs and the function of authorized participants

b. describe how ETFs are traded in secondary markets

c. describe sources of tracking error for ETFs

d. describe factors affecting ETF bid-ask spreads

e. describe sources of ETF premiums and discounts to NAV

f. describe costs of owning an ETF

g. describe types of ETF risk

h. identify and describe portfolio uses of ETFs

Exchange-Traded Funds

Page 1
LOS a
- explain

mutual fund shares/units —sell→ mutual fund
mutual fund shares/units ←buy— mutual fund

- creates and redeems all shares/units at closing NAV

- ETF shares → trade intraday on an exchange
→ are created/redeemed in-kind (shares-for-shares swap)
(created/redeemed: continuously)

Step 1: Primary market: OTC

works in reverse to redeem

Authorized Participant
- takes delivery to sell → called a creation unit (typically 50,000 shares)
- purchases to deliver (after close) → called a creation basket

ETF Sponsor —transfers ETF shares→ Authorized Participant
Authorized Participant —transfers securities→ ETF Sponsor

ETF Sponsor specifies & publishes each day the creation basket

Page 2
LOS a
- explain

AP incentive → arbitrage

e.g. 1/ ETF shares: $25.10 FV of creation basket: $25.00

buy low, sell high

- sell 50,000 shares at $25.10 for	$1,255,000	← cover short pos.
- Buy creation basket for:	1,250,000	→ deliver after close of market - get 50,000 ETF shares
gain	5,000	

e.g. 2/ ETF shares: $24.90 FV of creation basket: $25.00

- Buy 50,000 ETF shares for	(1,245,000)	→ deliver after market close
- Sell creation basket for	1,250,000	→ get redemption basket to cover short positions
gain	5,000	

Arbitrage gap: prices at which arbitrage makes sense (e.g. +/- .05)
- varies with the liquidity of the underlying

Page 3
LOS a
- explain

- dealer $\underbrace{\text{bid-ask spread}}$ on the ETF

determined by: cost of arbitrage
risk premium for volatility/liquidity risk

→ AP absorbs all costs of transacting the securities in the creation basket

∴ non-transacting ETF holders are shielded from the negative impact of these transaction costs (not with a MF)

- results in → ETF operates at lower costs
→ greater tax efficiency
→ ETF prices kept in-line with NAV

LOS b
- describe

⇒ Secondary market → investor to investor on an exchange
→ settlement is the same as other listed stocks (T + 2)

Note: APs may have up to 6 days settlement due to the time required to create/borrow ETF shares

Page 4
LOS c
- describe

- Expense Ratios: much lower than MFs
- no shareholder recordkeeping
- no communication costs
- no active research

broad-based, cap-weighted indexes
.03% - U.S. equities
.04% - U.S. bonds
.11% - emerging market equities

- ETF managers attempt to $\underbrace{\text{track an index}}$ as closely as possible (after fees)

typically evaluated using
$\underbrace{\text{daily return differences between ETF \& index}}$

s.d. of this difference series is called 'tracking error'

- typically for a 12-month period

- expectation is that an index fund would underperform its benchmark on an annual basis by the amount of its expense ratio

⇒ Sources of Tracking Error/

Page 5
LOS c
- describe

- Fees and expenses
- Representative sampling/optimization - full replication may not be optimal/possible for large indexes
  - esp. fixed-income

- can enhance or detract from ETF returns depending on whether holdings over/under-perform those in the index
- Depository receipts and ETFs - ETFs may hold shares that are different than those in the index
- Index changes - ETFs may add/eliminate constituents at different times/prices than the index
- Fund accounting practice - valuation times each day may differ from the index

⇒ Sources of Tracking Error/

Page 6
LOS c
- describe

- Regulatory & Tax Requirements e.g. foreign dividend withholding tax
- Asset manager operations - securities lending, foreign dividend recapture

⇒ Tax Issues

- Capital gains distribution - funds must distribute any capital gains realized during the year
  - usually low due to:

a) Tax Fairness · ETF investors sell to other ETF investors (does not require the fund to sell assets to meet redemptions)
· ETF shares are created/redeemed in-kind
→ not a taxable event

b) Tax efficiency - can use shares w/ largest unrealized gains in the redemption process
- cap. gains occur from - index reconstitution & rebalancing

- Other Distributions - dividends

⇒ ETF Trading Costs/ MF → acquire/redeem at end-of-day NAV

Page 7
LOS d
- describe

ETF → acquire/redeem any time exchange is open → closing price may include a premium or discount

→ commissions + trading costs — bid-ask spread / market impact

major impact from:
- market structure
- liquidity of underlying

US-Traded ETF Category	AUM ($ millions)	Average Spread ($ asset-weighted)	Median Spread
US Equity	1,871,942	0.03%	0.16%
International Equity	731,251	0.05%	0.24%
US Fixed Income	589,851	0.02%	0.14%
International Fixed Income	65,159	0.06%	0.24%
Commodities	62,620	0.05%	0.24%
Leveraged	29,633	0.29%	0.32%
Inverse	11,315	0.10%	0.21%
Asset Allocation	9,318	0.21%	0.29%
Alternatives	4,388	0.18%	0.38%
All US-Traded ETFs*	3,377,276	0.04%	0.20%

underlying securities trade in different markets
- makes it difficult to price simultaneously

use of futures

illiquid underlying

⇒ ETF Trading Costs/

Page 8
LOS d
- describe

bid-ask spread width factors:
- ETF liquidity (more volume = lower spreads)
- market maker competition (profit spread) - more comp. = lower spreads
- costs/risks for the MM (hedging risk, inventory)

$ETF_A \rightarrow Index_A \leftarrow F_A$ futures on the index = lower spread

+/ spreads tend to widen when market volatility increases or when information relating to the underlying index securities is expected

⇒ Premiums & Discounts/

Page 9
LOS e
- describe

premium or discount:

intraday price (ETF) vs. iNAV	close	closing price (ETF) vs. end-of-day NAV
- indicated NAV - intraday FV estimate based on its creation basket that day		- accurate assessment of FV (when underlying securities trade on same exchange) - same market structure, same closing time

Intraday:
Premium → ETF Price > iNAV
Discount → ETF Price < iNAV

End-of-day:
Premium → ETF Price > NAV
Discount → ETF Price < NAV

1/ Timing Differences:
- differences in closing times between the ETF & the underlying
Bonds: no true 'closing price' available
- may have stale pricing or/ model pricing may reflect what a dealer is willing to pay

⇒ Premiums & Discounts/

Page 10
LOS e
- describe

Bonds con't./ Bond ETFs may appear to trade at a $\underbrace{\text{premium}}_{\text{underlying model prices too low}}$ or $\underbrace{\text{discount}}_{\text{underlying model prices too high}}$

2/ Stale pricing → of the ETF itself

Equity	SPY	IVV	VOO	EUSA	IWM
Average spread (%)	0.00%	0.01%	0.01%	0.12%	0.01%
Average spread ($)	$0.01	$0.03	$0.03	$0.07	$0.01
Median premium/discount (%)[a]	0.00%	0.00%	0.00%	0.04%	0.01%
Maximum premium (%)[a]	0.12%	0.13%	0.18%	0.96%	0.12%
Maximum discount (%)[a]	–0.19%	–0.11%	–0.08%	–0.38%	–0.13%

Bonds	TLT	JNK	HYG
Average spread (%)	0.01%	0.03%	0.01%
Average spread ($)	$0.01	$0.01	$0.01
Median premium/discount (%)[a]	0.03%	0.10%	0.20%
Maximum premium (%)[a]	0.68%	0.41%	0.59%
Maximum discount (%)[a]	–0.52%	–0.67%	–0.75%

⇒ Costs of Ownership/

Page 11
LOS f
- describe

Fund Cost Factor	Function of Holding Period?	Explicit/ Implicit	ETFs	Mutual Funds
Management fee	Y	E	X (often less)	X
Tracking error	Y	I	X (often less than comparable index mutual funds)	(index funds only)
Commissions	N	E	X (some free)	
Bid–ask spread	N	I	X	
Premium/discount to NAV	N	I	X	
Portfolio turnover (from investor flows and fund management)	Y	I	X (often less)	X
Taxable gains/losses to investors	Y	E	X (often less)	X
Security lending	Y	I	X (often more)	X

Holding Period:	3 Months	12 Months	3 Years
Commission	0.10%	0.10%	0.10%
Bid–Ask Spread	0.15%	0.15%	0.15%
Management Fee	0.0375%	0.15%	0.45%
Total	**0.29%**	**0.40%**	**0.70%**
Trading Costs % of Total	0.86%	0.625%	0.36%
Management Fees % of Total	0.14%	0.375%	0.64%

Commission & Bid–Ask Spread: short-term
Management Fee → longer-term

⇒ Risks/

Page 12
LOS g
- describe

1/ Counterparty risk - some ETF structures involve reliance on a counterparty

e.g. ETNs - unsecured debt obligation of the issuer (a bank)

- settlement risk
  - when an ETF uses OTC derivatives
- securities lending → cash collateral minimizes this risk

2/ Fund Closures due to a) regulation
b) competition
c) corporate actions (M&A)
d) creation halts (mostly ETNs, arbitrage breaks down)
e) change in investment strategy

- can trigger capital gains and a need to find replacement investment

3/ Investor-Related Risk - lack of understanding of underlying exposures
- esp. leveraged & inverse

⇒ ETFs in Portfolio Management/

Page 13
LOS h
- identify
- describe

Portfolio Application	Role in Portfolio	Examples of ETFs by Benchmark Type
Cash Equitization/ Liquidity Management	Minimize cash drag by staying fully invested to benchmark exposure, transact small cash flows	Liquid ETFs benchmarked to asset category
Portfolio Rebalancing	Maintain exposure to target weights (asset classes, sub-asset classes)	Domestic equity, international equity, domestic fixed income
Portfolio Completion	Fill gaps in strategic exposure (countries, sectors, industries, themes, factors)	International small cap, Canada, bank loans, real assets, health care, technology, quality, ESG
Manager Transition Activity	Maintain interim benchmark exposure during manager transitions	ETFs benchmarked to new manager's target benchmark

⇒ ETFs in Portfolio Management/

Page 14
LOS h
- identify
- describe

Portfolio Application	Category	Role in Portfolio	Examples of ETFs by Benchmark Type
Core asset class or market	Strategic or tactical	Core long-term, strategic weighting Tactical tilt to enhance returns or modify risk Ease of access vs. buying underlying securities	Domestic equity, international equity, fixed income, commodities
Equity style, country, or sector; fixed income or commodity segment	Strategic or tactical	Tactical tilt to enhance returns or modify risk depending on short-term views Hedge index exposure of active stocks or bond strategy Ease of access vs. buying underlying securities	Value, growth, Japanese, Chinese, UK, Canadian, or Mexican equities; corporate or high-yield debt; gold; oil; agriculture
Equity sector, industry, investment theme	Dynamic or tactical	Tactical or dynamic active tilt to enhance returns or modify risk Efficient implementation of a thematic/ industry vs. single-stock view Capture performance on an emerging theme or innovation not reflected in industry categories	Technology, financials, oil and gas, biotech, infrastructure, robotics, gold mining, buybacks, internet innovation, cybersecurity

⇒ ETFs in Portfolio Management/

Page 15
LOS h
- identify
- describe

Portfolio Application	Category	Role in Portfolio	Examples of ETFs by Benchmark Type
Factor exposure	Strategic, dynamic, or tactical	Capture risk premium for one or more factors driving returns or risk Overweight or underweight depending on factor return or risk outlook Seek to capture alpha from rules-based screening and rebalancing (systematic active)	Quality, dividend growth, value, momentum, low volatility, liquidity screen, multi-factor
Risk management	Dynamic or tactical	Adjust equity beta, duration, credit, or currency risk	Currency-hedged, low-volatility, or downside-risk-managed ETFs
Leveraged and inverse exposure	Tactical	Access leveraged or short exposure for short-term tilts or risk management Limit losses on shorting to invested funds	ETFs representing asset classes, countries, or industries with leveraged or inverse daily return targets
Alternative weighting	Strategic, dynamic, or tactical	Seek outperformance from weighting based on one or more fundamental factors Balance or manage risk of security holdings	ETFs weighted by fundamentals, dividends, or risk; equal-weighted ETFs
Active strategies within an asset class	Strategic	Access discretionary active management in an ETF structure	ETFs from reputable fixed income or equity managers with active approach or theme
Dynamic asset allocation and multi-asset strategies	Dynamic or tactical	Seek returns from active allocation across asset classes or factors based on return or risk outlook Invest in a multi-asset-class strategy in single product	ETFs that allocate across asset categories or investment themes based on quantitative or fundamental factors

Using Multifactor Models

a. describe arbitrage pricing theory (APT), including its underlying assumptions and its relation to multifactor models

b. define arbitrage opportunity and determine whether an arbitrage opportunity exists

c. calculate the expected return on an asset given an asset's factor sensitivities and the factor risk premiums

d. describe and compare macroeconomic factor models, fundamental factor models, and statistical factor models

e. explain sources of active risk and interpret tracking risk and the information ratio

f. describe uses of multifactor models and interpret the output of analyses based on multifactor models

g. describe the potential benefits for investors in considering multiple risk dimensions when modeling asset returns

LOSs will match between the video and the PDFs, but may be in a different order than the CFAI readings

Using Multifactor Models

Page 1
LOS a-c
- describe
- define
- calculate

Factor → a variable or characteristic with which individual asset returns are correlated

- single factor model → CAPM

$$E(R_i) = R_F + \beta\underbrace{(R_M - R_F)}_{\text{ERP}}$$

ERP → risk factor return
- systematic or market risk

expected return (in equilibrium)

sensitivity to the market factor

- since asset specific risk can be diversified away, it should not be rewarded

- CAPM is an incomplete description of risk since anomalies exist

e.g. small-cap outperforms large-cap
value outperforms growth
momentum exists

∴ add more factors → multifactor model
- macro factors
- fundamental factors

Page 2
LOS a-c
- describe
- define
- calculate

→ Arbitrage Pricing Theory/

$$R_i = a_i + b_{i_1}F_1 + b_{i_2}F_2 + \cdots + b_{i_K}F_K + \varepsilon_i \longrightarrow \text{error term}$$

sensitivity to Factor i

Return to asset i in equilibrium

intercept
$E(R_i)$ given F = 0 $\forall F_i$

risk factor premiums
(i.e. return to Factor i)
corr$(F_{ij}) = 0 \; \forall F_{ij}$

or// $\underbrace{R_i - R_F}_{\text{excess return}} = a_i + b_{i_1}(F_1 - R_F) + b_{i_2}F_2 + \cdots + b_{i_K}F_K + \varepsilon$ (e.g. Carhart)

subtracts from one factor only

3 key assumptions/
1/ a factor model describes asset returns
2/ with many assets available, investors can form well-diversified portfolios that eliminate asset specific risk
3/ no arbitrage opportunities exist among well-diversified portfolios (a condition of financial market equilibrium)

Page 3
LOS a-c
- describe
- define
- calculate

→ Arbitrage Pricing Theory/

- if all assumptions hold → $E(R_P) = R_F + \beta_1 F_1 + \beta_2 F_2 + \cdots + \beta_K F_K$

(note: no α_P)

Ex. #1/#2

→ Carhart model/

$$R_P - R_F = a_P + b_{P1}RMRF + b_{P2}SML + b_{P3}HML + b_{P4}WML + \varepsilon_P$$

a_P → alpha

RMRF → return on a value-weighted index minus one-month T-Bill rate

SML → average return on 3 small-cap portfolios minus average return on 3 large-cap portfolios $(\beta_{SC} - \beta_{LC} = 0)$

HML → average return on 2 high $^{BV}/_{MV}$ portfolios minus average return on 2 low $^{BV}/_{MV}$ portfolios → no/little premium for mgmt., strategy, or business model $(\beta_H - \beta_L = 0)$

WML → past 12 month's winners (top 30%) minus past 12 month's losers (bottom 30%) $(\beta_W - \beta_L = 0)$

Page 4

→ Carhart model/ • since $E(a_i) = 0$

$$E(R_P) = R_F + \beta \bullet RMRF + \beta \bullet SML + \beta \bullet HML + \beta \bullet WML$$

→ Types of MF models/

LOS d
- describe
- compare

1/ macroeconomic factor models - factors are surprises in macroeconomic variables

(e.g. interest rates, inflation, business cycle risk, financial leverage)

- factors that affect CFs or discount rates

2/ fundamental factor models - attributes of stocks or companies

(e.g. $^{BV}/_{MV}$, market-cap, PE, financial leverage)

3/ statistical models - applied to historical returns to extract factors

e.g. portfolio of A, B, and C

F_1 = .1A + .6B + .3C
F_2 = .9A + .1B + .2C

• uncorrelated
• difficult to interpret

→ factor analysis
→ principal component analysis

Page 5
LOS d
- describe
- compare

→ Structure of fundamental factor models/

$$R_i = a_i + b_{i1}F_1 + b_{i2}F_2 + \cdots + b_{iK}F_K + \varepsilon_i$$

factors are returns

betas are standardized → calculated first

- except binary variables

$$b_{iK} = \frac{(\text{value of attribute k for asset i}) - (\text{Avg. value of attribute k})}{\sigma(\text{values of attribute k})}$$

$$\left(b_{iK} = \frac{X_i - \bar{X}}{\sigma}\right)$$

∴ stock with average value on attribute K will have $b_{iK} = 0$ and a value on attribute K 1 σ > average will have $b_{iK} = 1$

- factor sensitivities are calculated first, factor returns are then calculated through regression

Page 6
LOS d
- describe
- compare

→ Structure of fundamental factor models/

- factors can be:
  - company fundamental factors - related to internal performance (e.g. earnings growth/variability/momentum, fin. leverage)
  - company share-related factors - valuation measures, factors related to share price, trading characteristics) (e.g. $^{E}/_{P}$, $^{D}/_{P}$, $^{BV}/_{MV}$)
  - macroeconomic factors - sector/industry membership

→ Fixed-Income factor models/

- macroeconomic e.g. $R_i = a_i + b_{i1} \bullet F_{INFL} + b_{i2} \bullet F_{GDP} + \varepsilon_i$
- fundamental factors → Duration, Credit, Currency, Geography

e.g./ $R_i = a_i + b_{i1} \bullet F_{Gvt-sh} + b_{i2} \bullet F_{Gvt-Int} + b_{i3} \bullet F_{Gvt-Lt} + b_{i4} \bullet F_{Invest} + b_{i5} \bullet F_{HY-Y/d} + b_{i6} \bullet F_{MBS} + \varepsilon_i$

(all b_i estimated with a constrained regression such that $\sum b_i = 1$)

Ex. #3/4

Page 7
LOS d
- describe
- compare

→ Macroeconomic factor models/

$$R_i = a_i + b_{i1}F_1 + b_{i2}F_2 + \cdots + b_{iK}F_K + \varepsilon_i$$

ε_i: asset specific risk iff the model adequately represents the sources of common risk

a_i: $E(R_i)$ if all F = 0
i.e. expected return from expected macro factors

F: factors are surprises (actual - predicted)

- RP for GDP > 0
- RP for INFL < 0

$$E(R_i) = a_i + b_1 \cdot INFL + b_2 \cdot GDP$$ → inflation hedging ability

$$E(R_j) = a_i - b_1 \cdot INFL + b_2 \cdot GDP$$

- all else equal, req. ret. on i < req. ret. on j
- if INFL > 0 (act. - exp. > 0), $E(R_i)\uparrow$ and $E(R_j)\downarrow$

Ex. #5

Ex. #6

Page 8
LOS e, f
- explain
- describe
- interpret

- application of MF models
  - Return Attribution
  - Risk Attribution
  - Portfolio construction
  - Strategic portfolio decisions

→ Return Attribution/ - relative to a benchmark
- fundamental models are favoured
  - statistical models difficult to interpret
  - style choices better reflected vs. macro models

- attribute active return: $\underbrace{R_P - R_B}_{\text{sum of 2 components}}$

$$\sum_{k=1}^{K}(\beta_P - \beta_B)_k F_k + \text{Security Selection}$$

$(\beta_P - \beta_B)_k$: factor tilts; F_k: factor returns

Security Selection: $\left[(R_P - R_B) - \sum_{k=1}^{K}(\beta_P - \beta_B)_k F_k\right]$

Ex. #7

Page 9
LOS e, f
- explain
- describe
- interpret

→ Risk Attribution/ fundamental factor models typically used

Active risk = $\sigma_{R_A} = \sigma(R_P - R_B)$ = tracking risk/error (TE)

→ for absolute returns → SR = $\frac{R_P - R_F}{\sigma_P}$

→ for relative returns → IR = $\frac{R_P - R_B}{\sigma_{R_A}} = \frac{R_A}{\sigma_{R_A}}$

sum of 2 components

active factor risk (variance) + active specific risk or security selection risk (i.e. non-factor risk) (variance)

$$\left[= \sigma^2_{R_A} - \sum_{i=1}^{n} (W_i^a)^2 \, \sigma^2_{\sum i}\right] \qquad = \sum_{i=1}^{n} (W_i^a)^2 \, \sigma^2_{\sum i}$$

$(W_P - W_b)$

variance of returns not explained by the factors

Ex. #9

Page 10
LOS e, f
- explain
- describe
- interpret

→ Portfolio construction/

• passive mgmt. - replicate benchmark factor exposures on a much smaller set of securities

• active mgmt. - use MF models to predict α or construct a portfolio with a desired risk profile
(typically a pure factor portfolio)
b = 1 for that factor and b = 0 for all others)

• rules-based active mgmt. - overweight/underweight specific factors

Ex. #10

→ Strategic Portfolio decisions/

• MF models offer richer context over single factor models to improve portfolio selection

(R_F + risky asset portfolio)

• allows investors to obtain exposure to risks they are better suited to bearing

Page 11
LOS g
- describe

- build portfolios that replicate/modify the characteristics of an index in a desired way
- establish desired risk exposures to one or more risk factors
- perform granular risk/return attribution
  - ensure that active risk/return outcomes are commensurate with active fees
- help investors achieve better diversified portfolios & possibly more efficient portfolios

versus single-factor models

Measuring and Managing Market Risk

a. explain the use of value at risk (VaR) in measuring portfolio risk

b. compare the parametric (variance-covariance), historical simulation, and Monte Carlo simulation methods for estimating VaR

c. estimate and interpret VaR under the parametric, historical simulation, and Monte Carlo simulation methods

d. describe advantages and limitations of VaR

e. describe extensions of VaR

f. describe sensitivity risk measures and scenario risk measures and compare these measures to VaR

g. demonstrate how equity, fixed-income, and options exposure measures may be used in measuring and managing market risk and volatility risk

h. describe the use of sensitivity risk measures and scenario risk measures

i. describe advantages and limitations of sensitivity risk measures and scenario risk measures

j. explain constraints used in managing market risks, including risk budgeting, position limits, scenario limits, and stop-loss limits

k. explain how risk measures may be used in capital allocation decisions

l. describe risk measures used by banks, asset managers, pension funds, and insurers

Value-at-Risk

Page 1
LOS a
- explain

⇒ VaR – the ① minimum loss that would be expected
② • a certain %'age of the time
③ • over a certain time period
given the assumed market conditions

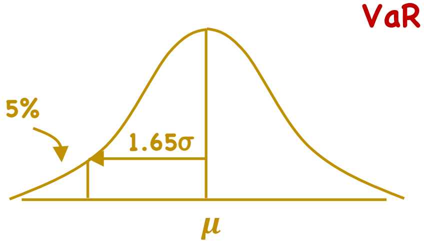
① expressed in either currency units or in percentage terms

e.g./ the 5% VaR of a portfolio is \$2.2M over a one-day period
– if P_V = \$400M, 5% VaR = 0.55% of P_V

a loss of at least \$2.2M is expected to occur once every 20 days (5% of the time)

VaR at 1% → 2.33σ
VaR at 1σ → 16%

time period could be:
weekly
monthly
quarterly



Estimating VaR

Page 2
LOS b, c
- compare
- estimate
- interpret

⇒ 3 methods to estimate VaR
- but first/
① convert the set of holdings in the portfolio into a set of exposures to risk factors
- risk decomposition
② gather a data history for each of the risk factors in the VaR model

then ③ estimate VaR
- parametric method
- historical simulation method
- Monte Carlo simulation method

running example/

	2 asset portfolio		P_V = \$150M
(S&P 500 ETF)		SPY	w_S = 80%
(L.T. Corporate Bond ETF)		LWC	w_L = 20%

• assumes the 2 securities are the risk factors

- return history of cash ETF → $\underbrace{\text{July 2012 - June 2014}}_{\text{lookback periods}}$

Page 3
LOS b, c
- compare
- estimate
- interpret

	Daily Avg. Ret.	Daily σ	Annualized Avg. Ret.	Annualized σ	subjective adjustment		
SPY	.08%	.70%	20.72%	11.14%	10.5%	20%	(σ)
LWC	.02%	.55%	4.21%	8.68%	6.0%	8.5%	(σ)

$\text{correlation}_{SPY,LWC} = -.20$

judgement suggests these may not be representative going forward

⇒ Parametric Method/ (variance–covariance method)

- begins with risk decomposition of the portfolio holdings
  - assumes return distributions for the risk factors are normal

⇒ Parametric Method/

Page 4
LOS b, c
- compare
- estimate
- interpret

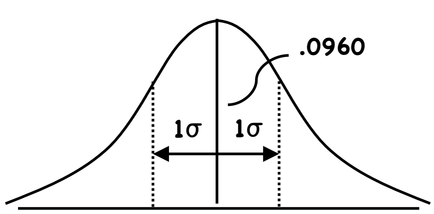
need $E(R_P)$ & σ_P

$$E(R_P) = w_S E(R_S) + w_L(R_L) = .80(.105) + .20(.06) = .096$$

$$\sigma_P = \sqrt{w_S^2\sigma_S^2 + w_L^2\sigma_L^2 + 2w_S w_L \rho_{SL}\sigma_S\sigma_L}$$

$$\underbrace{= \sqrt{.8^2(.2)^2 + .2^2(.085)^2 + 2(.8)(.2)(-.20)(.2)(.085)} = 0.157483}_{\text{inputs are based on annual returns}}$$

- daily conversions/



$$E(R_P) = {}^{.096}/_{250} = .000384$$

$$\sigma_P = {}^{.157483}/_{\sqrt{250}} = .009960$$

$$5\%\ VaR = [E(R_P) - 1.65\sigma_P](-1)P_V$$

$$= [.000384 - 1.65(.009960)](-1)150M$$

$$= \$2{,}407{,}500$$

Page 5
LOS b, c
- compare
- estimate
- interpret

⇒ Parametric Method/

- generally assumes distribution of returns on risk factors is normal (not an absolute requirement)

+/ simple & straightforward

-/ • VaR very sensitive to E(R) & σ

• difficult to use if $\underbrace{\text{the portfolio contains options}}_{\text{threatens normality}}$

⇒ Historical Simulation Method/

rather than

SPY	LWC
↓	↓
2 yrs.	2 yrs.
E(R)	E(R)
σ	σ

we use

$.8(SPY) + .2(LWC) = R_p$

↓

2 yrs.

Sort

-$

|

+$

Page 6
LOS b, c
- compare
- estimate
- interpret

⇒ Historical Simulation Method/

Comparison	Hist. Sim.	Param.
1%	2,314,998	3,423,462
5%	1,205,127	2,407,998
16%	618,287	1,441,800

- not constrained by an assumption of normality
- estimate VaR based on $\underbrace{\text{what actually happened}}_{\text{but no guarantee this will repeat}}$

∴ can handle any kind of financial instrument

⇒ Monte Carlo Simulation/

- not constrained by any distribution
- avoids the complexity of the parametric method when the portfolio has many risk factors

output ranked

-$

← 5%

+$

Advantages/Limitations

Page 7
LOS d
- describe

⇒ Advantages/

- simple concept
- Easily communicated concept
- Provides a basis for risk comparison (across asset classes, portfolios)
- facilitates capital allocation decisions
- can be used for performance evaluation (VaR can serve as the basis for risk adjustment to performance)
- reliability can be verified (backtesting)
- widely accepted by regulators

⇒ Limitations/

- subjectivity – VaR cutoff, time period, estimation method
- underestimating the frequency of extreme events
  - over reliance on the normal distribution

Page 8
LOS d
- describe

⇒ Limitations/

- failure to take into account liquidity
- sensitivity to correlation risk – in times of market stress, correlations among all assets tend to rise
- vulnerability to trending or volatility regimes
  - a portfolio may remain under its VaR limit every day, but just under it (losses would accumulate)
- misunderstanding the meaning of VaR
- oversimplification
- disregard of right-tail events

Extensions of VaR

Page 9
LOS e
– describe

- Conditional VaR – CVaR – (not technically a VaR measure)

  [avg. loss | VaR > cutoff] (typically obtained by backtesting)

  VaR > cutoff → min. loss at a certain confidence

  avg. loss → how much can I expect to lose if VaR is exceeded

  (a.k.a. expected tail loss, expected shortfall)

- Incremental VaR – IVaR – how VaR will change if a position size is changed relative to the remaining positions

e.g./ before	after
SPY w = 80%	SPY w = 90%
LWC w = 20%	LWC w = 10%
5% VaR = 2,407,530	5% VaR = $2,733,722

5% IVaR = 326,192

Page 10
LOS e
– describe

- Marginal VaR – MVaR – conceptually similar to IVaR, but reflects the effect of a very small change in the position

- Relative VaR – ex-ante tracking error – the degree to which the performance of a given portfolio might deviate from its benchmark

$$\underbrace{\text{(portfolio holdings − benchmark holdings)}}_{\text{active positions}}$$

Sensitivity & Scenario Measures

⇒ Sensitivity Risk Measures/ examines how performance responds to a single change in an underlying risk factor

Page 11
LOS f-i
- describe
- demonstrate

a) equity exposure measures – β (factor sensitivities)

(CAPM) $E(R_i) = R_F + \beta(E(R_M) - R_F)$

(multi-factor) $E(R_i) = R_F + \beta_{i1}F_1 + \beta_{i2}F_2 + \ldots + \beta_{iK}F_K$

b) fixed-income exposure measures – duration & convexity

$$\frac{\Delta B}{B} \approx -D\frac{\Delta y}{(1+y)} + {}^{1}/_{2}\,C\frac{\Delta y^2}{(1-y)^2}$$

c) option risk measures (non-linear payoffs)

$$\Delta(\text{delta}) \approx \frac{\text{change in value of option}}{\text{change in value of underlying}}$$

⇒ Sensitivity Risk Measures/

Page 12
LOS f-i
- describe
- demonstrate

c) option risk measures (non-linear payoff)

$$\text{gamma} \approx \frac{\text{change in } \Delta}{\text{change in value of underlying}}$$

$$\text{vega} \approx \frac{\text{change in value of option}}{\text{change in volatility of underlying}}$$

⇒ Scenario Risk Measures/ – estimates the portfolio return that would result from a hypothetical change in markets or a repeat of an historical event

- incorporates multiple factor movements (vs. single factor movements in sensitivity measures)
- use larger factor size movements

⇒ Scenario Risk Measures/

Page 13
LOS f-i
- describe
- demonstrate

a) historical scenarios – portfolio holdings are revalued first

- equities ⇒ can be modelled using their price histories as proxies for their expected behavior

- fixed-income & derivatives – may not have the relevant price history

- must be revalued based on what the inputs to a valuation model were historically

(re-priced under the conditions that prevailed during the scenario period)

chosen to represent extreme market dislocations (often exhibit abnormally high correlations among asset classes)

⇒ Scenario Risk Measures/

Page 14
LOS f-i
- describe
- demonstrate

a) historical scenarios

- scenario is run as if total price action across the whole scenario period happened instantaneously

output/ • total return
• total return vs. benchmark
• extra collateral/cash requirements as a result of the scenario

- modified approach ⇒ staggered scenario which allows expected management action (not too popular)

b) hypothetical scenarios (imagined, but have not yet been experienced)

- identify portfolios most significant exposures ⇒ reverse stress test

⇒ Scenario Risk Measures/

Page 15
LOS f-i
- describe
- demonstrate

b) hypothetical scenarios

stress test: event → factor risks → effect on portfolio

reverse stress test: factor risks → $\underbrace{\text{imagined events}}_{\text{what would increase the riskiness of the factor risks}}$ → effect on portfolio

- may also design 'rare, but not impossible' scenarios
  - earthquake, war, banking collapse
- goal is to understand the risks, not eliminate them
- if leverage is used, may use a single-factor stress test to assess if a given exposure is likely to impair their capital

⇒ versus VaR

Page 16
LOS f-i
- describe
- demonstrate

VaR – a measure of losses and the probability of large losses

S/S → changes in the value of an asset in response to changes in something else
- nothing about the probability of the change
- does inform about what is driving the risk

VaR – uses market returns from $\underbrace{\text{a lookback period}}_{\text{representative}}$

- Scenario – uses market returns from a specific unrepresentative time period

VaR, Sensitivity & Scenario – best used in combination

Page 17
LOS f-i
- describe
- demonstrate

⇒ Advantages/Limitations

+/
- they do not rely on history
- scenarios can be designed to overcome any assumption of normal distributions (can use 1, 10 or 1000 s.d. movements)
- scenarios can cheery pick which exposures get the worst movements

but/
- historical scenarios are not likely to repeat in exactly the same way
- hypothetical scenarios may incorrectly specify how assets will co-move
- hypothetical scenarios can be very difficult to create & maintain (all factors & their relationships)
- very difficult to know how to establish the appropriate limits (hypothetical)

Constraints

Page 18
LOS j ,k
- explain

- the limits placed on risk measurements
  - ↳ too high - constrains opportunity
  - ↳ too low - outsized losses

  } relates to risk appetite

1. Risk Budgeting - total risk appetite allocated to sub-activities

   e.g./ VaR limit = $4 million

market	credit	operational
$2M	$1.5M	.5M

2. Position limits - a control on overconcentration
   ↳ per issuer, country, currency, gross sizes
3. Scenario limits - a limit on the estimated loss for a given scenario
4. Stop-loss limits - when a loss of a particular size occurs in a specified period (reduce and liquidate)

4. Stop-loss limits

Page 19
LOS j ,k
- explain

e.g. 10-day 5% VaR = $5M, stop-less = $8M
cumulative monthly loss

- may hedge instead after losses of a given magnitude
  - called 'drawdown control' or 'portfolio insurance'

⇒ Risk Measures & Capital Allocation/
measurement of economic capital — capital needed to survive severe loss
(min. level established)

defines risk appetite in economic capital terms

sub divides (i.e. allocates capital)

units, portfolios, asset classes

• typically used when leverage or meaningful tail risk is present
(selling options, large concentrations)

Applications of Risk Measures

Page 20
LOS h
- describe

- 3 factors influence the types of risk measures used:
  1. degree of leverage - assess min. capitalization or max. leverage ratios
     high → stress tests, potential loss measures over very short periods
  2. mix of risk factor exposure - equity, fixed-income, options
  3. accounting/regulatory requirements

(economic capital: the amount of capital a firm needs to hold if it is to survive severe losses from the risks of its businesses)

FV (mutual funds, held-for-sale investments)
VaR, economic capital, duration, beta — rely on changes in FV

BV - asset/liability gap models

Page 21
LOS h
- describe

⇒ Banks

- liquidity gap – extent of any liquidity and asset/liability mismatch
- VaR – for the 'held-for-sale/trading' portion of the B.S.
- Leverage – weight assets by risk, riskier assets assigned greater weight, then divide by equity
  (more equity required to support riskier assets)
- Sensitivities – 'held-for-sale' – duration/key-rate/partial for interest rate portion
  - forex exposure, commodity exposure, equity exposure
  - delta, gamma & vega for options
- economic capital – applied to full balance sheet
  - blends market, credit & operational risks
  - total loss at high (99.9%) level of confidence (1 yr.)
- scenario analysis – stress tests applied to whole balance sheet
  - assess capital sufficiency

Page 22
LOS h
- describe

⇒ Asset Managers/

- not regulated in terms of capital or liquidity requirements
- risk mgmt. measures focused primarily on volatility, probability of loss, probability of underperforming a benchmark

a) Traditional asset managers (use little/low/no leverage)

- risk measures include:
  - position limits (country, currency, sector, asset class) %age of portfolio value
  - sensitivities
  - Beta sensitivity (for equity-only accounts)
  - liquidity (%age of ADV/security)
  - scenario analysis
  - active share – measure of %age of portfolio that differs from the benchmark

Page 23
LOS h
- describe

⇒ Asset Managers/

a) Traditional asset managers

- redemption risk – open-end funds
- ex-post vs. ex-ante TE

$(R_P - R_B)$ historical — assess potential TE based on current P vs. Benchmark

- VaR – more for absolute return strategies

b) Hedge Funds (use of leverage)

- sensitivities – full range
- gross exposure long, short & gross (|long| + |short|)
- leverage
- VaR – high confidence intervals (> 90%), short time periods
- Scenarios – specific risk stress tests

Page 24
LOS h
- describe

⇒ Asset Managers/

b) Hedge Funds

- Drawdown – worst returning month or quarter
  – worst peak-to-trough decline

⇒ Pension Funds/ (defined benefit)

- interest rate & curve risk • provides the discount rate for arriving at the PV of the liability
- surplus at risk

liability driven investing

assets – long positions
liabilities – as short fixed income positions

measure how much assets might underperform the liabilities (over the year)

– if some threshold level is surpassed, changes may be made to asset allocations

⇒ Pension Funds/

Page 25
LOS h
- describe

- glide path - multi-year stages to change a portfolio from a current state to a target state
- liability hedging exposures vs. return generating exposures

⇒ Insurers/ (significant regulation & accounting oversight)

how they must retain reserves and reflect their liabilities

property & casualty

- sensitivities & exposures
- economic capital & VaR
- scenario analysis

market risks in the portfolio
characteristics of the insurance exposure

⇒ Insurers/

Page 26
LOS h
- describe

life insurers/ stronger ties to financial markets

- sensitivities
- asset/liability matching
- scenario analysis

Backtesting and Simulation

a. describe objectives in backtesting an investment strategy

b. describe and contrast steps and procedures in backtesting an investment strategy

c. interpret metrics and visuals reported in a backtest of an investment strategy

d. identify problems in a backtest of an investment strategy

e. evaluate and interpret a historical scenario analysis

f. contrast Monte Carlo and historical simulation approaches

g. explain inputs and decisions in simulation and interpret a simulation

h. demonstrate the use of sensitivity analysis

Backtesting and Simulation

LOS a (1p)	Objectives of Backtesting – describe
LOS b, c (15.5p)	Backtesting Process – describe, contrast, interpret
LOS d (10p)	Common Problems in Backtesting – identify
LOS e (4p)	Historical Scenario Analysis – evaluate, interpret
LOS f (6p)	Simulation Analysis – contrast
LOS g (4p)	Monte Carlo Simulation – explain
LOS h (5p)	Sensitivity Analysis – demonstrate

Page 1
LOS a
– describe

Objective → assess the viability of a strategy using historical data

– fits well with quantitative and systematic investment styles, also used by fundamental managers

– implicit assumption → the future will somewhat resemble the past

---

LOS b, c
– describe
– contrast
– interpret

Step #1: Specify investment hypothesis and goal(s)

- investment hypothesis: a method (trading rule, security selection criteria) aimed at achieving the goal
- goal(s):
  - beat a benchmark
  - max. SR

Determine investment rules and process

e.g./ Taylor rule

if:	output gap	infl. gap	
	neg.	neg.	→ ow. small-cap growth
	pos.	pos.	→ ow. large-cap growth
	pos.	neg.	→ neutral
	neg.	pos.	→ neutral

Page 2
LOS b, c
- describe
- contrast
- interpret

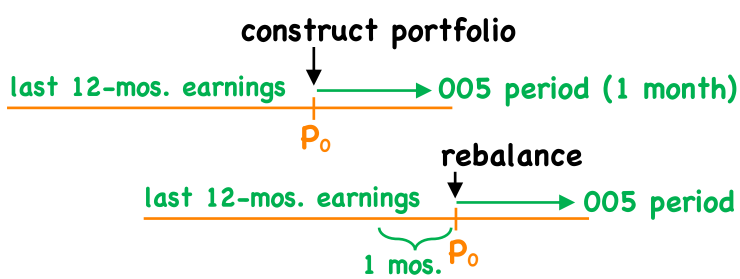
Step #1:
Decide key parameters

- Investment universe → the securities available for investment (e.g. Russell 3000)
- Return definition - what currency will the return be computed
  - no fx hedging → all returns in the home currency
  - complete hedging → all returns in local currency
  - if the goal is excess return, benchmark must be specified (typically related to the investment universe)
- Rebalancing/Reconstitution frequency and transactions costs
  - typically monthly
  - more frequently increases transaction costs
- Start and End dates - as long as possible, but does risk multiple regimes (Ex. #1)

Page 3
LOS b, c
- describe
- contrast
- interpret

Step #2: Historical Investment Simulation

- construct portfolios to be tested
  - depends on the investment hypothesis
    - may be an identified portfolio of securities
    - may be a trading strategy
- ensure that it is rebalanced based on a pre-determined frequency
  - rolling windows typically used



(exh. #2)

Page 4
LOS b, c
- describe
- contrast
- interpret

Step #3: Analysis of Backtesting Output

- average return
- risk profiles – Sharpe ratio, Sortino ratio, volatility, maximum drawdown
- visual output – returns vs. normal distribution
  – cumulative performance vs. index or benchmark

Ex. #2

→ Backtesting Multifactor Models/ – most quantitative selection models use a multifactor structure

Step #1: Strategy Design/

1/ Benchmark factor portfolio → equal weights each factor

$.25F_1 + .25F_2 + .25F_3 + .25F_4$

all stocks equally weighted →

$.25F_1$	$.25F_2$	$.25F_3$	$.25F_4$	
long	long	long	long	→ top 20%
short	short	short	short	→ bottom 20%

pure factor portfolios

Page 5
LOS b, c
- describe
- contrast
- interpret

→ Backtesting Multifactor Models/

1/ Benchmark factor portfolio

e.g./ Defensive value → trailing $^{E}/_{P}$; Growth → $^{FYI}/_{FYO}$ growth ; Leverage $^{D}/_{E}$

2/ Risk parity portfolio – $\frac{\text{Total } \sigma_P}{N_F}$ → contribution to σ_P of each factor portfolio

- requires a complete variance-covariance matrix at each rebalancing date

Step 2: Historical Investment Simulation

- rolling window procedure implemented twice
  - once at the factor level (rebalancing/reconstitution)
  - again at factor portfolio level (rebalancing)

Page 6
LOS b, c
- describe
- contrast
- interpret

→ Backtesting Multifactor Models/
Step 2: Historical Investment Simulation

BM - rebalance back to equal weight
RP - at each month end, a var.-covar. matrix must be re-estimated to rebalance back to risk parity

Step 3: Output analysis - Exh. 7/8/9

LOS d
- identify

1/ Survivorship Bias - index constituents today represent those companies that have survived to today
- should use point-in-time data

e.g./ constituents of Russell 3000 in 1995, not 2021

- rebalancing/reconstitution will, over time, eliminate those that disappear

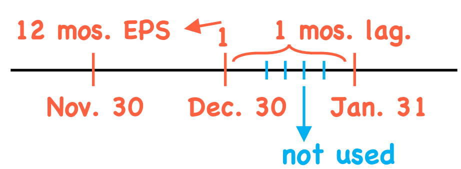
---

Page 7
LOS d
- identify

2/ Look-ahead bias - using information that was unknown or unavailable during the historical period

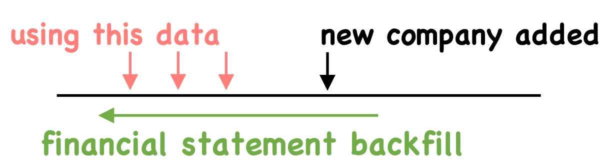
- types: • reporting lags → e.g. 12-mos. lagging EPS as of Dec. 31/2000
- Q4/2000 would not have been available until 3-10 weeks later

- can use a lag:



Ex. #12

• data revisions
- revised data was not available at the time
- should use original released value
- new additions to a database



Page 8
LOS d
- identify

3/ Data snooping → running multiple tests looking for significance

- testing at 95%, $^{5}/_{100}$ tests will be significant by chance alone

- can be mitigated by raising the critical t-value or performing cross-validation

---

LOS e
- evaluate

→ Historical scenario analysis - a type of backtesting that explores the performance of an investment strategy in different structural regimes and at structural breaks

e.g./ • from expansions to recessions
• from low volatility to high volatility regimes

linear deterministic rolling-window backtest

exh. #14-16

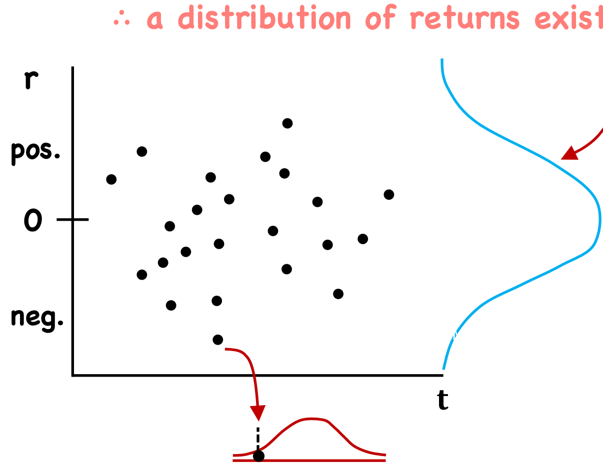
Page 9
LOS f
- contrast

→ Simulation analysis - 2 basic types of simulation

1/ historical simulation - past returns are selected at random from many different historical periods

- however, only one set of realized data exists

∴ a distribution of returns exist over time, but only one point in time



Page 10
LOS f
- contrast

2/ Monte Carlo simulation - each variable is assigned a statistical distribution and observations are drawn at random
- can specify any distribution

- verifies performance of backtesting by accounting for randomness in time

→ 8 steps to a simulation:
1/ define target variable → portfolio return
2/ specify key decision variables → returns on each underlying asset or factor and their weights
3/ specify the number of trials (1,000 to 10,000)
4/ define distributional properties of the key decision variables - Monte Carlo only
5/ use a random number generator to draw N random numbers
6/ compute the value of the target variable for each set of input variables

Page 11
LOS f
- contrast

→ 8 steps to a simulation:
7/ repeat N times
8/ calculate metrics

LOS g
- explain

→ Historical Simulation/
- sampling with replacement or without (bootstrapping)
- most common method used in investment research

• Process - using BM_P/RP_P

1. target variables = BM_r and RP_r
2. key decision variables = returns on each factor
3. N = 1000 trials
4. Bootstrapped sampling - draw a random # from 0 < x < 1
   - based on x, observe factor returns for that month

interval width = $1 / \text{\# of periods}$ month = $X / \text{interval width}$

Page 12
LOS g
- explain

- Process - using BM_P/RP_P

5. iterate 1000 times with replacement to get all factor returns
6. weight each factor return to get BM_r/RP_r
7. Repeat for all 1000 months

equal — use weights from final month of period

8. calculate performance metrics, plot the distribution of returns

Exh. 17-20

→ Monte Carlo simulation/ follows similar steps as historical simulation but does require a distribution description for each key decision variable

- critical → use of regression and distribution-fitting techniques to estimate parameters (mean, sd, skew, kurtosis)
- called model calibration
- fMultivar in R, Python, Matlab → simplify this step

Page 13
LOS g
- explain

→ Monte Carlo simulation/

- distribution should reasonably describe the pattern of the underlying data
- correlations between key decision variables are required

- bias-variance trade-off → more complexity, low specification errors, high estimation errors

- for simplicity, asset classes or factors are used in simulations

30 securities	5 factors
30 μ, 30 sd, 435 cov.	5 μ, 5 sd, 10 cov.

- distribution of asset or factor returns modelled as a multivariate normal distribution (means, sds, covars.)
- does not account for deviations (Skew, K) (Ex. #7)

$$X \sim N_3(\mu, \Sigma) \qquad \mu = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \qquad \begin{bmatrix} \text{var.} & \text{cov.} & \text{cov.} \\ \text{cov.} & \text{var.} & \text{co.} \\ \text{cov.} & \text{cov.} & \text{var.} \end{bmatrix}$$

$$X_1 \sim N(1, \text{var.})$$
$$X_2 \sim N(2, \text{var.})$$
$$X_1 + X_2 + X_3 \sim N(6, \text{var.})$$

Page 14
LOS h
- demonstrate

→ Sensitivity Analysis/ - explores how a target variable is affected by changes in input variables

- the multivariate normal distribution fails to account for deviations from normality

∴ can test a multivariate skewed Student's t-distribution

exh. #24/25

REVIEW

Economics & Investment Markets

Review - 1

$$P_t^i = \sum_{s=1}^{N} \frac{E_t(\widetilde{CF}_{t+s})}{(1 + I_{t,s} + \theta_{t,s} + \rho_{t,s})^s}$$

I - real risk-free rate

θ - expected inflation

ρ - risk premium associated with $\widetilde{CF}$

$\widetilde{CF}$ - uncertain cash flows

$E_t[\widetilde{CF}_{t+s}]$ - expectation of CF conditional on information available today

- marginal utility of consumption

today vs. tomorrow

intertemporal rate of substitution

$$\tilde{m}_{t,s} = \frac{MU_{t+s}}{MU_t} \Rightarrow I$$

- good economic times more wealth today
- lower MU_t

expectation of:

high wealth at t+1 low MU_{t+1} - higher I required to defer consumption

low wealth t+1 high MU_{t+1} - lower I required to defer consumption

Review - 2

I & $\tilde{m}_{t,s}$ are inversely related
- high $\tilde{m}_{t,s}$, low I
- low $\tilde{m}_{t,s}$, high I

- higher levels of wealth
  - higher levels of I (wealth lowers the price of safe assets)
    - more willing to buy risky assets

- multi-period premiums/

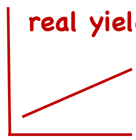
future asset price	$\tilde{m}_{t+s}$	
high	low	negative covariance
low	high	

- implies lower price, which implies higher I as time increases (i.e. upward sloping real yield curve)
- increase in real GDP growth → increase in real rates (I)
- higher trend real GDP growth - higher I than an economy with lower trend real GDP growth
- increase in real GDP growth volatility → increase in real rates (I)

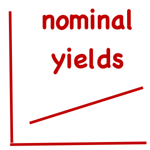
Review – 3

$\theta_{t,s} + \pi_{t,s}$ – premium for inflation (θ) + uncertainty in the forecast of inflation (π)

BEI – break-even inflation rate: nominal yield (default-free zero)
– real yield (default-free zero)
= BEI



– driven by $\widetilde{m}_{t,s}$



– driven by $\widetilde{m}_{t,s}, \theta, \& \ \pi_{t,s}$

– upward sloping curve may imply
- future rate increases
- risk premiums
- rate decreases + larger risk prem.

⇒ Credit Premiums/
- credit spreads smaller in expansions, wider in recessions
- spreads of lower rated bonds are more volatile (versus higher rated bonds)
  - contract more in good times
  - widen more in bad times

Review – 4

⇒ Credit Premiums/ – some sectors are more sensitive to the business cycle than others

cyclical vs. non-cyclical

$$P_t^i = \sum_{s=1}^{N} \frac{E_t[\widetilde{CF}_{t+s}]}{1 + I_{t,s} + \theta_{t,s} + \pi_{t,s} + \gamma_{t,s}^i}$$

credit risk

- company specific factors
  - profitable, low debt interest pmts., low debt levels etc…

– sovereign credit risk (emerging/developing economies)

⇒ Equity Risk Premium/

$$P_t^i = \sum_{s=1}^{N} \frac{E_t[\widetilde{CF}_{t+s}]}{1 + I_{t,s} + \theta_{t,s} + \pi_{t,s} + \underbrace{\gamma_{t,s}^i + \kappa_{t,s}^i}_{\lambda_{t,s}^i \Rightarrow \text{ERP}}}$$

additional return required to invest in equities over and above credit risky bonds
- compensates for poor hedging properties of equities for bad economic times

Review - 5

- Valuation Multiples/
    - higher forward P/E
        - an ↑ in $E[\widetilde{CF}]$
        - a decrease in I, θ, π, γ, κ
    - value tends to outperform growth in the aftermath of a recession
    - growth tends to outperform value when economy is expanding
    - small cap - larger ERP, large cap - smaller ERP

- Commercial Real Estate/

$$P_t = \sum_{s=1}^{\infty} \frac{\overbrace{E_t[\widetilde{CF}_{t+s}]}^{\text{low sensitivity to business cycle}}}{1 + I_{t,s} + \theta_{t,s} + \pi_{t,s} + \gamma_{t,s} + \kappa_{t,s} + \underbrace{\phi_{t,s}}_{\text{illiquidity}}}$$

- property value - high sensitivity to business cycle

Analysis of Active Management

Review - 1

active return/

$R_A = R_P - R_B$ vs. $\alpha = R_p - \beta_P R_B$

$$\sum_{i=1}^{N} \Delta w_i R_{A_i} = \sum_{i=1}^{N} \Delta w_i (R_i - R_B)$$

B = benchmark:
1) representative of assets to be selected
2) replicable at low cost
3) weights verifiable ex ante, return data available ex post

- positive value is added by: overweight securities if $R_i > R_B$; underweight securities if $R_i < R_B$

$$= \underbrace{\sum_{j=1}^{M} \Delta w_j R_{B_j}}_{\text{asset allocation}} + \underbrace{\sum_{j=1}^{M} w_{P,j} R_{A_j}}_{\text{security selection}}$$

total value added (M = # of asset classes)

= Σ(active weight × benchmark return)

+ Σ(absolute portfolio weight × active return)

Sharpe ratio $SR_p = \dfrac{R_P - R_F}{\sigma R_P}$ - unaffected by the addition of cash/leverage

Information ratio $IR = \dfrac{R_P - R_B}{\sigma(R_p - R_B)} = \dfrac{R_A}{\sigma R_A}$ $\dfrac{\text{active return}}{\text{active risk}}$

- affected by cash/leverage

- unaffected by the aggressiveness of the active weights

Review - 2

$SR_p^2 = SR_B^2 + IR^2$ - portfolio with the highest IR will also have the highest SR

- optimal active risk:

$$\sigma_{R_A} = \frac{IR}{SR_B} \cdot \sigma_{R_B}$$

- constrained: subject to a limit on active risk

- optimal active weights

$$\therefore \Delta w_i^* = \frac{\mu_i}{\sigma_i^2} \cdot \frac{\sigma_A}{K\sqrt{BR}} = N$$

and: $\mu_i = IC\sigma_i S_i$ - the investor's subjective expectations of R_{A_i}

μ_i - forecasted active returns

Portfolio Construction: (ability to put ideas into action) Transfer Coefficient (TR) $\left(\text{Corr } \frac{\mu_i}{\sigma_i}, \Delta w_i \cdot \sigma_i\right)$

risk adjusted

Signal Quality: Information Coefficient (IC) $\left(\text{Corr } \frac{\mu_i}{\sigma_i}, \frac{R_{A,i}}{\sigma_i}\right)$ (forecasting ability)

w_i - active weights

$w_i \longleftrightarrow R_{A,i}$: value added $\left(\text{Corr } \frac{\Delta w_i}{\sigma_i}, \frac{R_{A,i}}{\sigma_i}\right)$

$R_{A,i}$ - realized active returns

(Page 17)

Page 17
LOS c
- state
- interpret

e.g./

Security	Score	Volatility
1	1.0	25%
2	1.0	50%
3	-1.0	25%
4	-1.0	50%

Securities 1 and 2: outperform
Securities 3 and 4: underperform

assume IC = .20

① Forecasted Active Returns/ $E(R_{Ai}) = \mu_i$

$\mu_i = IC\sigma_i S_i$

1 .20 (.25) (1) = 5%
2 .20 (.50) (1) = 10%
3 .20 (.25) (-1) = -5%
4 .20 (.50) (-1) = -10%

② Active returns are uncorrelated
- independent forecasts are made each yr.

∴ BR = 4

③ Max. $E(R_A)$ s.t. $\sigma_A = 9.0\%$

- calculate Δw_i^*

$$\therefore \Delta w_i^* = \frac{\mu_i}{\sigma_i^2} \cdot \frac{\sigma_A}{IC\sqrt{BR}}$$

$$\Delta w_1 = \left(.05/.25^2\right)\left(.09/.20\sqrt{4}\right) = 0.18$$

$$\Delta w_2 = \left(.10/.5^2\right).225 = .09$$

$$\Delta w_3 = \left(-.05/.25^2\right).225 = -.18$$

$$\Delta w_4 = \left(-.10/.5^2\right).225 = -.09$$

$$\sum = 0$$

Review - 3

- Basic Fundamental Law/ $E(R_A)^* = IC\sqrt{BR}\,\sigma_A = IR^*\,\sigma_A$

(* indicates use of optimal Δw_i)

$(IR^* = IC\sqrt{BR})$

optimal expected active return is a positive function of
- forecasting ability
- breadth
- active risk

(pg. 18 & 19)

BR - the # of independent decisions made/yr.

- Full Fundamental Law/ $E(R_A) = \underbrace{(TC)(IC)\sqrt{BR}}_{IR}\,\sigma_A$

(TC) 1. The greater the ability to use optimal active weights, the higher the expected return

(IC) 2. The higher the IC (i.e. better forecasting ability), the higher the expected return

$\sqrt{BR}$ 3. The more a manager can re-visit allocation decisions, the higher the BR, the higher the expected return

σ_A 4. The higher σ_A, the higher $E(R_A)$

⇒ The Basic Fundamental Law/

Page 18
LOS c
- state
- interpret

$$E(R_A)^* = IC\sqrt{BR}\,\sigma_A$$

* indicates the use of optimal active security weights

- optimal expected active return is a positive function of
  - forecasting ability (IC)
  - breadth
  - active risk

Recall: $IR = \frac{E(R_A)}{\sigma_A}$

so… $IR^* = \frac{IC\sqrt{BR}\,\sigma_A}{\sigma_A} = IC\sqrt{BR}$ thus $E(R_A)^* = IR^*\,\sigma_A$

e.g./ revisited

Security	Δw_i	μ_i	w_{B_i}	w_{P_i}	R_{P_i}
1	.18	.05	.25	.43	.15
2	.09	.10	.25	.34	.20
3	-.18	-.05	.25	.07	.05
4	-.09	-.10	.25	.16	.0

$$E(R_P) = .43(.15) + .34\,(.20) + .07\,(.05) + .16\,(0) = 13.6\%$$

$$E(R_A) = 13.6\% - 10\% = 3.6\%$$

$$E(R_A) = .18(.05) + .09(.10) - .18\,(-.05) - .09(-.10) = 3.6\%$$

⇒ The Basic Fundamental Law/

Page 19
LOS c
- state
- interpret

e.g./ revisited

Security	Δw_i	μ_i	w_{B_i}	w_{P_i}	R_{P_i}
1	.18	.05	.25	.43	.15
2	.09	.10	.25	.34	.20
3	-.18	-.05	.25	.07	.05
4	-.09	-.10	.25	.16	.0

$E(R_P) = 13.6\%$
$E(R_A) = 3.6\%$

$$\sigma^2_{R_A} = (.18)^2(.25)^2 + (.09)^2(.50)^2 + (-.18)^2(.25)^2 + (-.09)^2(.5)^2$$

$$\sigma_{R_A} = \sqrt{.002025 + .002025 + .002025 + .002025} = .09$$

$\sigma_A = 9\%$

- using Basic Fundamental Law

$$E(R_A)^* = IC\sqrt{BR}\,\sigma_A = IR^*\,\sigma_A$$

$$= .20\sqrt{4}\,(.09) = .40(.09) = 3.6\%$$

Review - 3

- **Basic Fundamental Law/** $E(R_A)^* = IC\sqrt{BR}\,\sigma_A = IR^*\,\sigma_A$

(* indicates use of optimal Δw_i)

optimal expected active return is a positive function of
- forecasting ability
- breadth
- active risk

$(IR^* = IC\sqrt{BR})$

BR – the # of independent decisions made/yr.

(pg. 18 & 19)

- **Full Fundamental Law/** $E(R_A) = \underbrace{(TC)(IC)\sqrt{BR}}_{IR}\,\sigma_A$

(TC) 1. The greater the ability to use optimal active weights, the higher the expected return

(IC) 2. The higher the IC (i.e. better forecasting ability), the higher the expected return

$\sqrt{BR}$ 3. The more a manager can re-visit allocation decisions, the higher the BR, the higher the expected return

σ_A 4. The higher σ_A, the higher $E(R_A)$ (pg. 21)

---

e.g./

Page 21

LOS c
- state
- interpret

Security	$E(R_{A_i})$	σ_{A_i}	Δw_i^*	Δw_i
1	5%	25%	18%	6%
2	10%	50%	9%	4%
3	-5%	25%	-18%	7%
4	-10%	50%	-9%	17%

$\underbrace{\sigma_A = .09}_{\text{active risk constraint}}$

① $TC = CORR\left(\frac{\mu_i}{\sigma_i}, \Delta w_i \sigma_i\right) \longrightarrow$

Pairs

$.05/.25 = .20$	&	$.06(.25) = 0.015$
$.10/.50 = .20$		$.04(.50) = 0.02$
$-.05/.25 = -.20$		$.07(.25) = 0.0175$
$-.10/.50 = -.20$		$-.17(.50) = -0.085$

② $E(R_A)^* = 3.6\%\quad \sigma_A^* = 9.0\%$

$E(R_A) = .06(.05) + .04(.10) + .07(-.05) - .17(-.10) = 2.1\%$

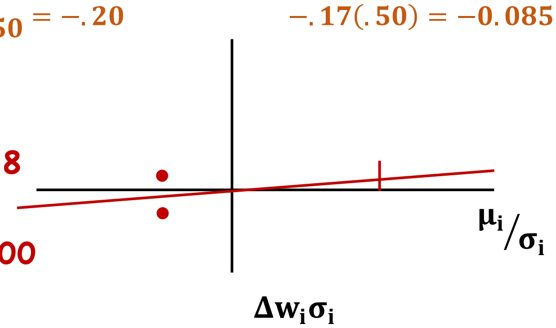
$\sigma_A = 9.0\%$

③ $E(R_A) = TC \cdot IC \cdot \sqrt{BR} \cdot \sigma_A$

$= (.58)(.20)(\sqrt{4})(.09)$

$= 2.088\%$

TC = .58
vs
TC* = 1.00



Review - 4

E(IR) – single best criterion for assessing active performance among managers with the same benchmark

- for any given asset class, choose the manager with the highest IR

⇒ Adjusting risk by combining the active portfolio with the benchmark

$$\sigma_{R_A} = TC \cdot \frac{IR^*}{SR_B} \cdot \sigma_{R_B} \qquad SR_P^2 = SR_B^2 + (TC)^2(IR^*)^2$$

(pg. 23)

⇒ Ex-Post Performance/

$$E(R_A | IC_R) = (TC)\,(IC_R)\sqrt{BR}\,\sigma R_A$$

IC_R – realized forecasting ability

constraints

realized R_A = $\underbrace{E(R_A \mid IC_R)}_{\text{skill – may be pos. or neg.}}$ + noise → may be pos. or neg.

Page 23
LOS c, d
- state
- interpret
- explain

$$\sigma_{R_A} = TC \, \frac{IR^*}{SR_B} \cdot \sigma_{R_B} \qquad SR_P^2 = SR_B^2 + (TC)^2(IR^*)^2$$

e.g./ $IR^* = .30$ $\quad SR_B = 0.40$ $\quad \sigma_{R,B} = .16$ $\quad TC = .50$

optimal amount of aggressiveness/

$$\sigma_{R_A} = (.50)\frac{.30}{.40}(.16) = 6\%$$

then $SR_P = \sqrt{(64)^2 + (.5)^2(.3)^2} = 0.43$

- if the constrained portfolio had $\sigma R_A = .08$

$$w_B = 1 - \frac{\text{optimal } \sigma_A}{\text{actual } \sigma_A} = 1 - {.06}/{.08} = .25$$

$$w_P = .75$$

Review - 4

E(IR) – single best criterion for assessing active performance among managers with the same benchmark

- for any given asset class, choose the manager with the highest IR

⇒ Adjusting risk by combining the active portfolio with the benchmark

$$\sigma_{R_A} = TC \cdot \frac{IR^*}{SR_B} \cdot \sigma_{R_B} \qquad SR_P^2 = SR_B^2 + (TC)^2(IR^*)^2$$

(pg. 23)

⇒ Ex-Post Performance/

$$E(R_A | IC_R) = \underbrace{(TC)}_{\text{constraints}} (IC_R)\sqrt{BR}\,\sigma R_A$$

IC_R – realized forecasting ability

$$\text{realized } R_A = \underbrace{E(R_A \mid IC_R)}_{\text{skill – may be pos. or neg.}} + \text{noise} \rightarrow \text{may be pos. or neg.}$$

ETFs

Review – 1

LOS a/ created/redeemed in kind

works in reverse as well

Authorized Participant ← transfers ETF shares ← Sponsor

Authorized Participant → transfers securities (after market close) → Sponsor

↓ Sells ETF shares

↑ Buy securities

creation basket published daily

• Arbitrage gap → prices at which arbitrage makes sense

- AP absorbs all costs of transacting in the securities

LOS b/ secondary market → investor to investor
→ same settlement process as stocks (T + 2)

Review – 2

LOS c/ • expense ratios → lower than MFs
→ ETF managers attempt to track an index as closely as possible

- tracking difference = daily return difference between index and fund
- tracking error = s.d. of tracking difference (annualized)

Sources of TE:

- fees/expenses
- representative sampling/optimization – full replication may not be optimal
- use of ADRs/ETFs
- Index changes – timing differences
- Fund accounting practice – valuation times
- Regulatory & Tax Requirements – for. div. withholding
- Asset manager operations – sec. lending, for. div. recapture

Review – 3

LOS d/ +/–
- • creation/redemption fees, trading costs
- \+ bid-ask spreads of underlying securities
- \+ compensation for risk of hedging or carrying positions
- \+ MM profit spread (competition)
- \- Discount related to likelihood of receiving an offsetting ETF order in a short time frame (liquidity)

LOS e/ iNAV – indicated NAV (intraday) → Premium: ETF Price > iNAV else discount
NAV – end of day NAV → Premium: ETF Price > NAV, else discount

1/ Timing differences → closing times between markets
2/ Stale pricing → illiquid ETF or illiquid underlyings

Review – 4

LOS f/

		f(Holding Period)	
• management fees		Y	→ lower preferred by long-term holders
• Bid-ask spread	trading costs	N	→ lower preferred by short-term traders
• Commission	trading costs	N	→ lower preferred by short-term traders

LOS g/ 1/ Counterparty risk
- settlement risk
- securities lending

2/ Fund closures
- regulation
- competition
- corporate actions
- creation halts
- change in investment strategy (soft closures)

3/ Investor-related risks
- lack of understanding, expectation of performance

Review - 5

LOS h/ 1/ Efficient Portfolio Management
- cash equitization
- rebalancing
- portfolio completion
- manager transition

2/ Asset Class Exposure
- core exposure → asset class/sub-class
- tactical strategies → style, sector, theme

3/ Active & Factor Investing
- factor exposure
- risk management - adjust β, duration
- levered/inverse exposure
- Alternative weighting → fundamental
- Active strategies (in an ETF structure)

Using Multifactor Models

Review - 1

→ LOS a-c - describe/define/calculate/

- single factor model
$$E(R_P) = R_F + \beta(R_M - R_F)$$

- multi-factor model
$$E(R_P) = R_F + \underbrace{\beta_{P_1}}_{\text{factor sensitivities}} F_1 + \beta_{P_2} F_2 + \cdots + \beta_{P_K} \underbrace{F_K}_{\text{factors}}$$

- Arbitrage pricing theory: 3 key assumptions
  - diversification eliminates asset specific risk
  - no arbitrage among well-diversified portfolios
  - a factor model describes asset returns

- Carhart 4-factor model

$$\underbrace{R_P - R_F}_{\text{excess return}} = a_P + b_1 \bullet \underbrace{RMRF}_{\text{equity risk premium (market)}} + b_2 \bullet \underbrace{SML}_{\text{small - large (size)}} + b_3 \bullet \underbrace{HML}_{\text{high } BV/MV \text{ - low } BV/MV \text{ (value)}} + b_4 \bullet \underbrace{WML}_{\text{winners - losers (momentum)}} + \varepsilon_P$$

Review - 2

→ LOS d - describe/compare/ types of MF models

1/ fundamental factor models

$$R_i = a_i + \underbrace{b_1}_{\text{betas are standardized}} F_1 + b_2 F_2 + \cdots + b_K \underbrace{F_K}_{\text{factors are returns}}$$

$$b_i = \frac{(\text{value of attribute k for asset i}) - (\text{Avg. value of attribute k})}{\sigma(\text{values of attribute k})}$$

- factors can be: company fundamental factors (earnings, CF, leverage)
  company share-related factors (valuation, share price, trading)
  macroeconomic factors (sector/industry)

→ Fixed Income: macroeconomic (inflation, GDP growth)
  fundamental (duration, credit, currency)

Review – 3

→ LOS d – describe/compare/ types of MF models

2/ macroeconomic factor models

$$R_i = a_i + b_1F_1 + b_2F_2 + \cdots + b_KF_K + \varepsilon_i$$

• asset specific risk iff the model adequately represents the sources of common risk

$E(R_i)$ of all F = 0

∴ a_i reflects E(R) of predicted values

factors are surprises (actual – expected)

– assumes returns are correlated with surprises

so: R_i = $\underbrace{\text{expected return}}_{\substack{a \\ \text{predicted}}}$ + $\underbrace{\text{unexpected return}}_{\substack{(b_kF_k) \\ \text{actual – predicted}}}$ + $\underbrace{\text{asset specific return}}_{\substack{\varepsilon \\ \text{unexplained by factors}}}$

3/ statistical factor models
- factor analysis
- principal component analysis

– factors are weighted combinations of the securities in the portfolio

Review – 4

LOS e, f – explain/describe/interpret/

1/ Return attribution – typically fundamental models used

Active Return = $R_P - R_B$

$\sum_{k=1}^{K} (\beta_P - \beta_B)_k F_k$

factor tilts — $(\beta_P - \beta_B)_k$

factor returns — F_k

Security selection

Active return – factor return

2/ Risk Attribution – fundamental models typically used

Active risk = $\sigma_{R_A} = \sigma(R_P - R_B)$

active factor risk

security selection risk

$(\text{Active risk})^2 - (\text{security selection risk})^2$

$\sum_{i=1}^{n} (W_i^a)^2 \, \sigma^2_{\Sigma i}$ — variance of returns not explained by the factors

W_i^a = $(W_P - W_b)$

Review - 5

LOS e, f - explain/describe/interpret/

3/ Portfolio construction

- passive mgmt. - replicate benchmark factor exposures
- active mgmt. - predict α or construct portfolios with a desired risk profile

e.g. pure factor portfolio, b = 1 with b = 0 on all other factors

- rules-based active mgmt. - ow/uw specific factors

4/ Strategic portfolio decisions

- more choice than just risk-free + risky asset
- engineer risk exposures better suited to holding

Review - 6

LOS g - describe/

- replicate/modify characteristics of an index
- establish desired risk exposures
- perform return/risk attribution
- build better diversified and more efficient portfolios

Measuring/Managing Market Risk

Review - 1

- VaR - the minimum loss
  a certain %age of the time
  over a certain time period

  given current market conditions

- 3 methods to estimate VaR
  - parametric method
  - historical simulation
  - Monte Carlo simulation

Step #1: convert holdings in portfolio into a set of exposures to risk factors
Step #2: gather a data history for each of the risk factor
Step #3: estimate VaR

differ by method

① Parametric Method (variance-covariance method)

- need $E(R_p)$ & σ_p

$$E(R_P) = w_{F_1} \cdot F_1 + w_{F_2} \cdot F_2 \ldots$$

$$\sigma_P = \sqrt{w_{F_1}^2\sigma_{F_1}^2 + w_{F_2}^2\sigma_{F_2}^2 + 2w_{F_1}w_{F_2}\sigma_{F_1}\sigma_{F_2}\rho_{F_1F_2}}$$

- if daily VaR

$$E(R_P)/_{250}$$

$$\sigma_P/_{\sqrt{250}}$$

$$5\%\ \text{VaR} = [E(R_P) - 1.65\sigma_P](-1)P_V$$

$$1\%\ \text{VaR} = [E(R_P) - 2.33\sigma_P](-1)P_V$$

Review - 2

① Parametric Method - best used when confident that normality can be reasonably approximated and parameter estimates are reliable (esp. covariances)
- sensitive to estimates of $E(R_p)$ & σ_p
- difficult to use if the portfolio contains options

② Historical Simulation Method
- reprice current portfolio 'as if' it existed for some lookback period (complex if n is large)
- run against historical risk factor data
- rank returns from lowest to highest and take desired %'age percentile value

- best used when the distributions of factor returns during the look back period are expected to be representative
- not constrained by any assumption of normality ∴ can handle any financial instrument
- estimate VaR on what actually happened

Review - 3

③ Monte Carlo Simulation - not constrained by any distribution
- inputs are the risk factors
- rank output lowest to highest, take desired %'age percentile value

Advantages of VaR/ • simple concept, basis for risk comparison, facilitates capital allocation decisions, widely accepted by regulators

Limitations of VaR/ Subjectivity (VaR cut-off, lookback period, estimation method)
• underestimating the frequency of extreme events, failure to take account of liquidity, misunderstanding the meaning of VaR, a portfolio may remain under VaR - but just under it.

Conditional VaR/ [avg. loss | VaR > cutoff]
- expected tail loss, expected shortfall

Marginal VaR/ change in VaR for a 1-unit change on each holding

Relative VaR/ ex-ante tracking error (portfolio holdings - benchmark holdings)

Review - 4

Incremental VaR/ how VaR changes if weightings in the portfolio change

e.g./ before: w_1 = 80% w_2 = 20% after: w_1 = 90% w_2 = 10%
5% VaR = X 5% VaR = Y IVaR = $\underline{Y}$ - $\underline{X}$

- Sensitivity Risk Measures/ how performance responds to a single change in an underlying risk factor

Equity/	Fixed Income/	Options/	
β	• Duration	• Delta	• Theta
	• Convexity	• Gamma	• Rho
		• Vega	

- Scenario Risk Measures/ - hypothetical change in markets or a repeat of an historical event
a) historical scenario - portfolio holdings are revalued first
equities → price history
bonds/derivatives → revalued based on what the inputs to a valuation model were historically
i.e. current tenor, but historical rates, volatility

- Scenario Risk Measures/

Review - 5

a) historical scenarios

- scenario is run as if total price action across whole scenario period happened instantaneously

• total return • total return vs. benchmark • collateral/cash req.

- modified approach - staggered scenario to allow mgmt. response

stress test → identified event → factor risks → effect on portfolio

b) hypothetical scenarios - imagined, but have not yet been experienced (or rare, but not impossible)

- goal is to understand risk, not eliminate it

reverse stress test factor risks → imagined events → effect on portfolio

- model multiple factor movements, larger size of each factor movement

Backtesting and Simulation

Review - 1

LOS a - describe/ assess the viability of a strategy using historical data (quant. + fundamental managers)

- implicit assumption → future will somewhat resemble the past

LOS b, c - describe, contrast, interpret/

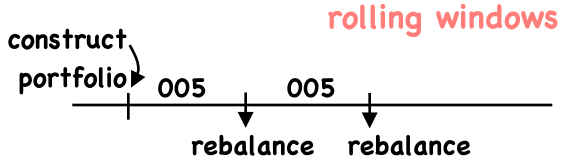
Step #1: • Specify investment hypothesis and goals

a method aimed at a goal — relative or absolute return

• Determine investment rules and process
• Decide key parameters (Investment universe, return definition, rebalancing/reconstitution frequency, start/end dates)

Step #2: Historical Investment Simulation - construct portfolios to be tested, ensure rebalancing is done on schedule

rolling windows typically used



Review - 2

LOS b, c - describe, contrast, interpret/

Step #3: Analysis of Backtesting Output

- average return
- risk profiles - SR, Sortino ratio, sd, max. DD
- visual output - return vs. normal distribution

LOS d - identify/

1/ Survivorship bias - using index constituents that did not exist at some point in the historical period (use point-in-time data)

2/ Look-ahead bias:
- using information (fundamental or economic) that was unknown or unavailable during the historical period
  - can use reporting lags for fundamental data
  - use original released data, not revisions

3/ Data snooping: running multiple tests on a data set looking for significance (raise critical-t, use cross-validation)

Review - 3

LOS e - evaluate/ • Historical scenario analysis - a type of backtesting that explores the performance of an investment strategy in different structural regimes and at structural breaks

- still a linear deterministic rolling-window backtest

LOS f - contrast/

1/ historical simulation - past returns are selected at random from many different historical periods

- may use the same data points as backtesting, just not linear

2/ Monte Carlo simulation - each variable is assigned a statistical distribution and observations are drawn at random → does not use single-point historical returns

∴ accounts for randomness in time

Review - 4

LOS g - explain/ Steps:

1/ define target variable - portfolio return

2/ specify key decision variables → asset class or factor returns + weights

3/ specify # of trials

4/ define distributional properties of key decision variables (Monte Carlo)

- correlations required
- joint distribution assumption required - multivariate normal
  - multivariate skewed Student's t-dist.

• historical → sampling method
↓
bootstrapping = with replacement

5/ Draw N random numbers

6/ compute target variable

7/ repeat N times

8/ produce output

LOS h - demonstrate/ Sensitivity analysis - explores how a target variable is affected by changes in input variables